Learning to Select Actions for Complex Hand Tasks: A Comparative Study of Visuomotor Policies and Latent World Models

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[PLACEHOLDER: Submission Date]

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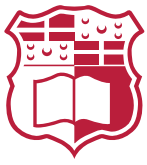

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I declare that the use of artificial intelligence tools in the preparation of this dissertation was limited to writing assistance, code suggestions, and literature search support. All research design, experimental decisions, implementation choices, result interpretation, and academic claims are my own.

Signature: _____

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List of Abbreviations

ACT Action Chunking Transformer.

ALOHA A Low-cost Open-source Hardware System for Bimanual Teleoperation.

BC Behaviour Cloning.

CDiT Conditional Diffusion Transformer.

CEMP Cross-Entropy Method with Policy.

CNN Convolutional Neural Network.

CVAE Conditional Variational Autoencoder.

DDIM Denoising Diffusion Implicit Models.

DDPM Denoising Diffusion Probabilistic Models.

DexWM Dexterous Manipulation World Model.

DiT Diffusion Transformer.

DP Diffusion Policy.

GATO Generalist Agent.

HDF5 Hierarchical Data Format 5.

I-JEPA Image Joint Embedding Predictive Architecture.

JEPA Joint Embedding Predictive Architecture.

LeWM LeWorldModel.

RMSE Root Mean Square Error.

SSL Self-Supervised Learning.

V-JEPA2 Video Joint Embedding Predictive Architecture 2.

V-JEPA2-AC Action-Conditioned Video Joint Embedding Predictive Architecture 2.

VAE Variational Autoencoder.

ViT Vision Transformer.

VLM Vision Language Model.

1 Background

This chapter assumes familiarity with the basic mechanics of a neural network and introduces six concepts central to this work: visuomotor policies, rigid-body pose in $\mathrm{SE}(3)$, latent representations, conditional VAEs, transformer attention, and self-supervised learning. These foundations support understanding of the specific architectures in Chapter 2 and underpin the candidate selection mechanism and world model evaluator design examined in Chapter 3.

1.1 Visuomotor policies

A visuomotor policy is a learned function that takes a visual observation as input and produces an action or sequence of future actions as output. In a manipulation setting, the visual observation is typically a single camera frame showing the hand and the object being manipulated, and the action describes how the hand should move next. The policy is called visuomotor because it connects vision directly to motor output, without requiring a hand-crafted intermediate representation of what the scene contains [1].

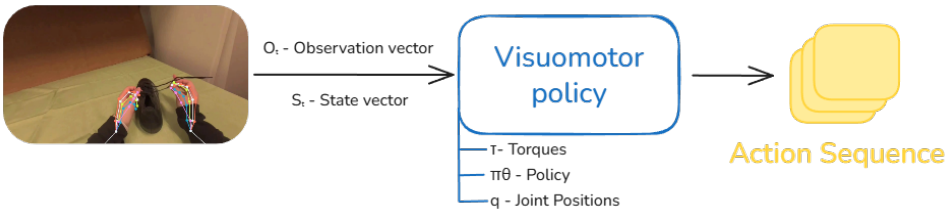


Figure 1.1 A visuomotor policy maps visual observations to motor actions. The input is a camera frame showing the hand and object, and the output is a proposed action for how the hand should move next.

Learning this mapping from data is appealing because useful hand behaviour depends on subtle visual cues that are difficult to specify by hand. The timing of a reaching movement, the grip shape adopted before contact, and the postural adjustments made in response to object geometry all depend on information visible in the camera image but difficult to encode as explicit rules [2–4]. A visuomotor policy learns these associations directly from recorded demonstrations, capturing the relationship between what the camera sees and what the hand does without requiring the programmer to describe it explicitly.

1.1.1 Behaviour cloning and compounding error

The most direct way to train a visuomotor policy from demonstrations is Behaviour Cloning (BC) [1]. Here an observed frame predicts the action taken by the expert at that moment, and the policy is trained to minimise the difference between its predicted action and the recorded expert action across all frames in the dataset [5]. The problem arises when a small prediction error causes the hand to move to a state that is slightly different from any state in the training data. This results in a compounding error: as the policy encounters increasingly unfamiliar states, its prediction errors grow larger over time.

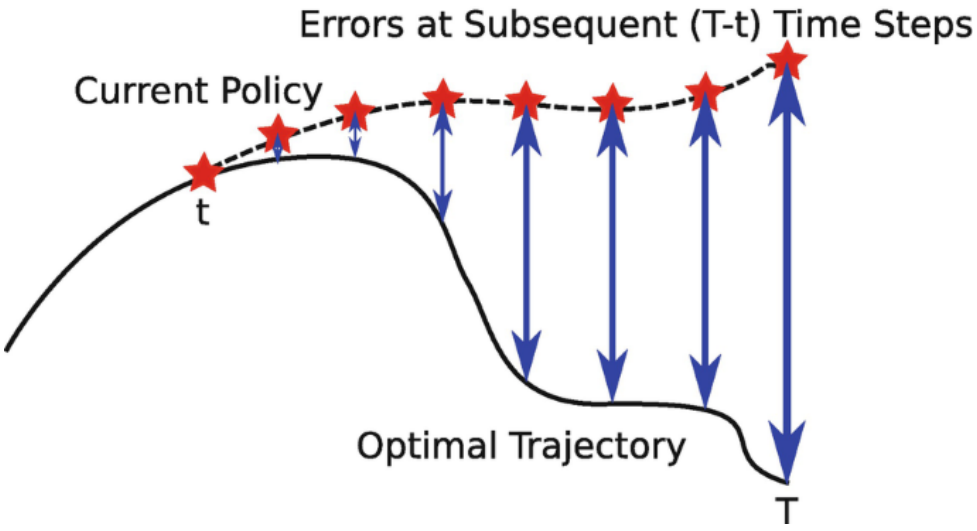


Figure 1.2 Compounding error in behaviour cloning. The policy trajectory drifts progressively further away at each step. The growing gap between the two paths illustrates how small errors can accumulate over time [6].

1.1.2 Action chunking

Action chunking addresses the compounding error problem by changing what the policy predicts. Instead of predicting a single next action at each step, the policy predicts a short sequence of future actions in one forward pass. This sequence is called an action chunk [7, 8]. This is why Figure 1.1 shows the action sequence as a stack of multiple action blocks. By committing to a chunk of k future steps, where k is a hyperparameter set per task, the policy makes one prediction decision every k frames rather than one per frame. With fewer decision points, there are fewer opportunities for errors to accumulate, and the hand stays closer to the training distribution for longer [9].

1.2 Rigid body pose and the $\mathrm{SE}(3)$ representation

To predict how a hand will move, a system needs a precise way to describe where the hand is and which way it is pointing. These two quantities together are called the pose of a rigid body. For the human wrist, pose is fully described by its position in space and its orientation relative to a fixed reference frame [10].

Translation – Position is represented as a vector of three numbers, one for each spatial dimension: how far the wrist is to the right or left, how far forward or backward, and how far up or down [2]. This vector $\mathbf{t} = [t_x,\ t_y,\ t_z]^\top$ is straightforward to predict with no special constraints on the output.

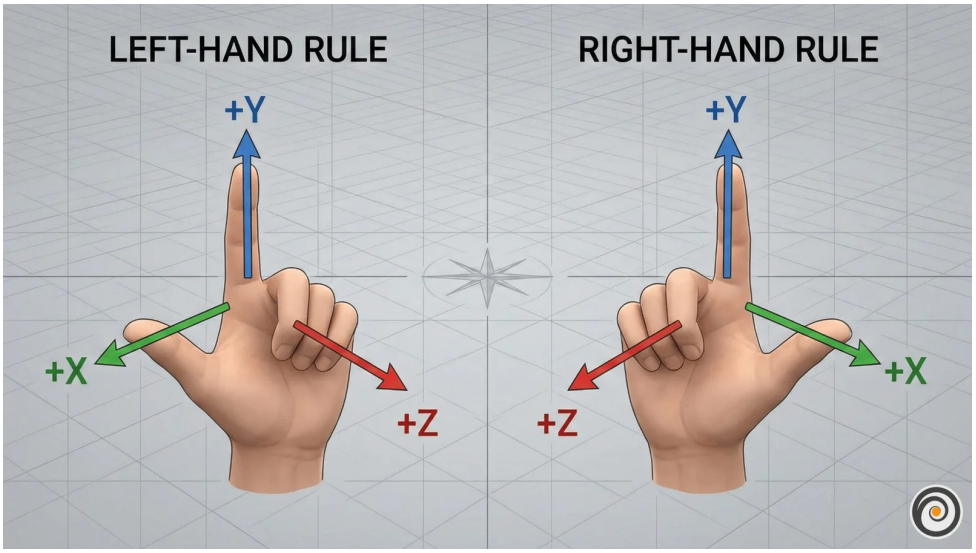


Figure 1.3 Translation for each arm can be represented by a movement along each of the three spatial axis [11].

Rotation – Orientation requires more care. A common representation uses three angles known as Euler angles, one per axis. These are intuitive but problematic for learning. At certain configurations the axes align and the system loses a degree of freedom, a singularity called gimbal lock. A subtler problem is discontinuity: a wrist rotated 359 degrees and one rotated 1 degree are almost physically identical, yet their angle values are numerically far apart. A network minimising prediction error will produce large errors near these boundaries even when the true motion is smooth [12].

A rotation matrix R avoids both issues: small physical rotations produce small numerical changes in the matrix entries, and there are no singularities.

The homogeneous transform – Translation and rotation are combined into a single 4×4 matrix, the standard representation of a pose in the special Euclidean group $\mathrm{SE}(3)$:

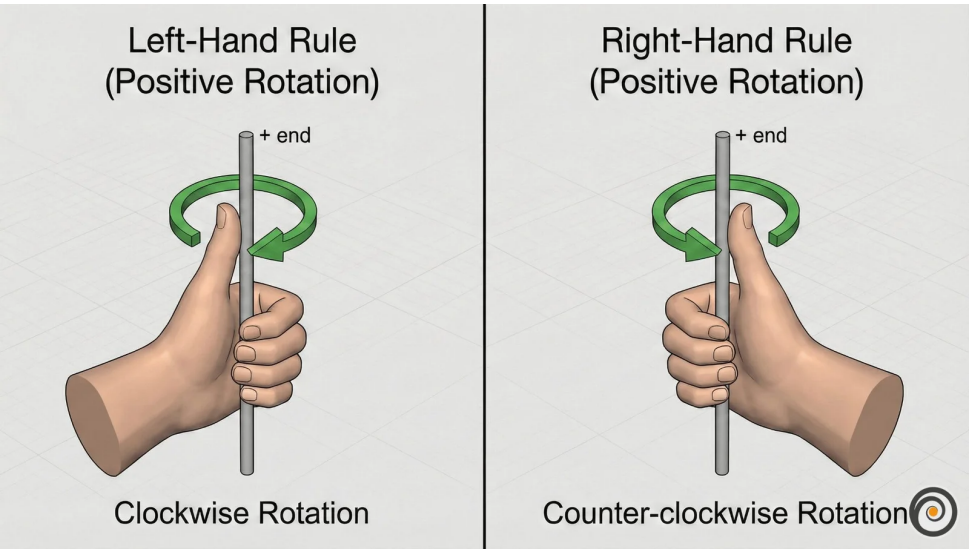
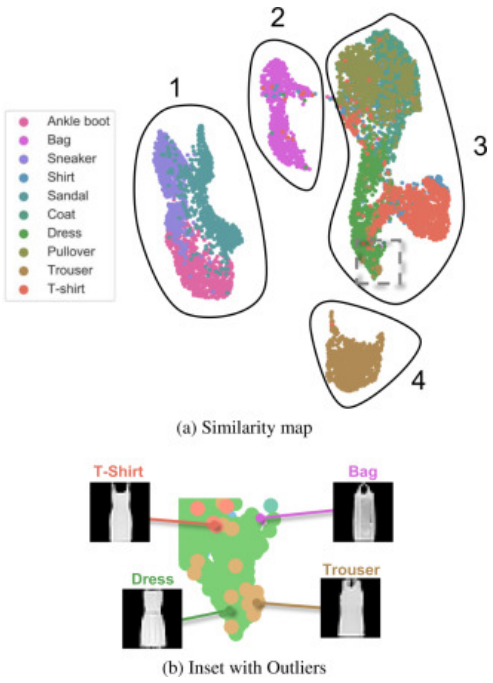


Figure 1.4 Rotation around the y-axis for each arm. The same circular movement can be imagined around the x and z axes; the total orientation of the wrist is the combined result of rotations around all three [11].

$$T = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \tag{1.1}$$

1.3 Latent representations

A latent representation is a learned internal description of an observation. An image starts as a grid of pixel values, where each pixel stores three or four channel intensities (typically red, green, and blue, with an optional transparency value). A model does not usually reason with these raw numbers directly. Instead, it transforms them through several layers into a new set of values that no longer correspond to individual pixel coordinates or colours. These new values form a compact encoding that captures patterns in the image such as shape, position, motion, and spatial structure. The collection of these encodings is often called the latent space. Figure 1.5 shows an example in which similar items cluster together in the learned representation.



(a) Similarity map

(b) Inset with Outliers

Figure 1.5 Example latent-space plot showing how structurally similar items cluster together in the learned representation. Structural similarity here refers to items that share shape, geometry, and likely physical behaviour [13].

The purpose of this transformation is to keep task-relevant structure while discarding unnecessary visual detail. For a manipulation scene, a useful latent state might encode hand position, object pose, contact progression, or motion trend without preserving every texture or lighting variation. A latent state is therefore not directly interpretable in the way an image is, but it can still preserve the information needed to reason about what happens next.

This is why latent prediction is often a better fit than pixel-level prediction for high-dimensional video observations. Predicting the next frame at the pixel level requires a model to account for every visual detail of the scene, including lighting changes, texture variation, and background movement that carry no information about task-relevant dynamics. A latent predictor instead operates on compact embeddings and concentrates predictive capacity on the structure that determines what actually happens next [14]. This design trade-off motivates the world model architectures reviewed in Chapter 2.

1.4 Conditional variational autoencoders

A VAE maps an input to a distribution of latent vectors described by a mean μ and standard deviation σ, rather than a single point. Sampling different latent vectors lets the decoder produce different but plausible outputs rather than one fixed answer. Figure 1.6a shows this standard encoder-latent-decoder flow [15].

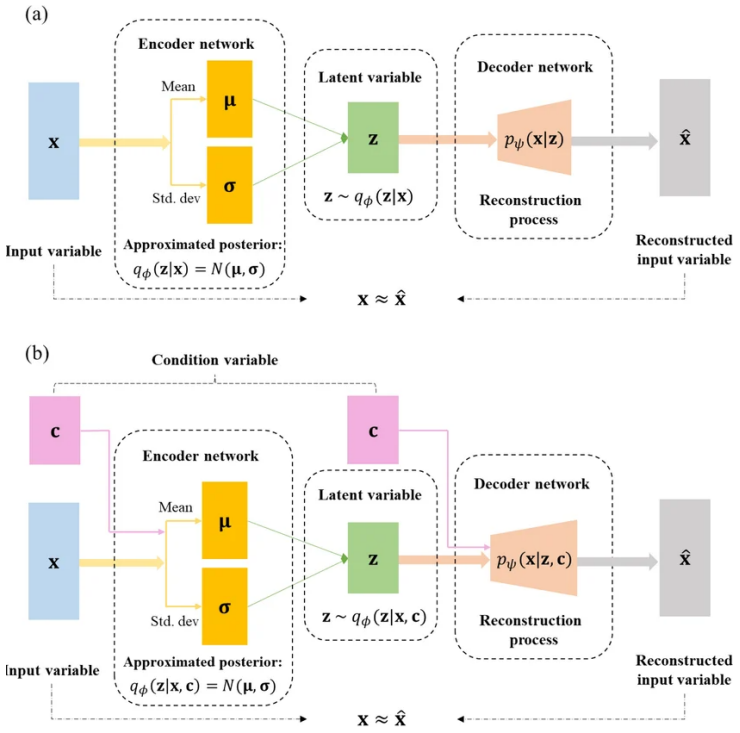


Figure 1.6 Standard VAE (a) and conditional CVAE (b) encoder-decoder pipelines [16].

For the policy to be trainable end-to-end, gradients must flow from the action prediction loss all the way back to the encoder. Sampling z directly from a distribution blocks this path. The reparameterisation trick resolves this by writing $z = \mu + \sigma\epsilon$, where ϵ is standard normal noise drawn independently of the learned parameters. Figure 1.7 shows the full encoder-latent-decoder pipeline alongside its training loss: a reconstruction term that rewards accurate output and a regularisation term that keeps the latent distribution close to a standard Gaussian, together forming the evidence lower bound [15].

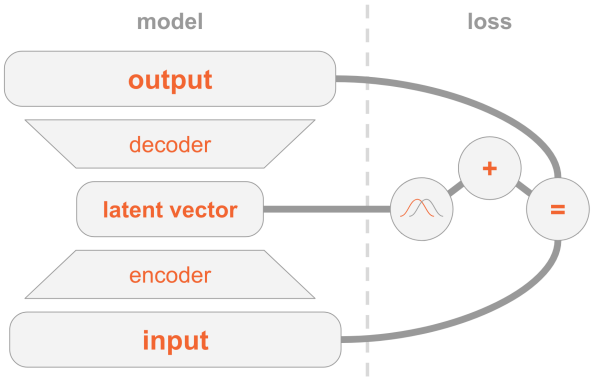


Figure 1.7 VAE encoder-decoder architecture and two-part training loss (reconstruction and regularisation) [17].

A CVAE adds an extra conditioning input to both encoder and decoder, shown in Figure 1.6b. In a visuomotor policy, this condition is the current observation frame,

so the latent variable only needs to capture how the future action varies within that scene [18]. The ACT architecture uses this to sample multiple latent vectors from one observation and decode each into a candidate action chunk [7].

1.5 Transformer networks and the attention mechanism

A transformer processes a sequence as a set of tokens. A token can represent a word, an image patch, a timestep in an action sequence, or any vector produced by an earlier model component. The architectural idea needed here is simple: each token is updated by looking across the other tokens and deciding which of them are most relevant. This relevance-weighted exchange of information is called attention [19].

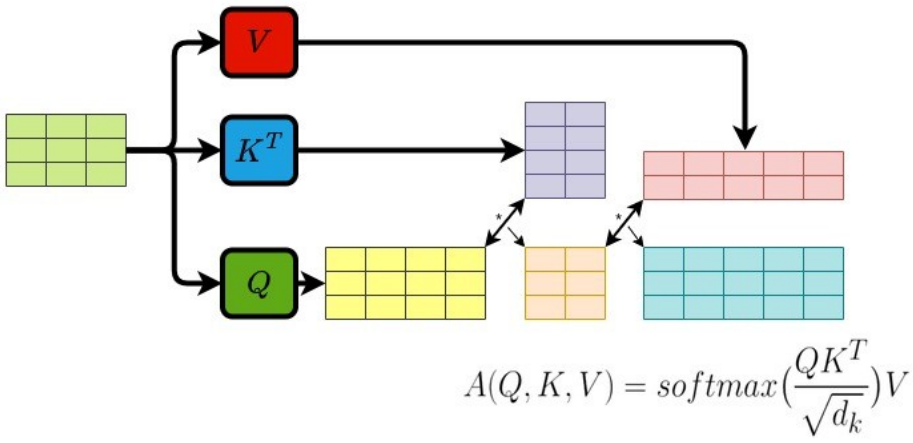


Figure 1.8 Transformer attention block: each token queries all others with Q, gathers weighted values V via keys K, and the weighted sum yields the updated output token [20].

Attention forms three learned views of every token:

- **Query** (Q) – what this token is looking for.
- **Key** (K) – what another token contains.
- **Value** (V) – what another token contributes if attended to.

Figure 1.8 shows how each token (highlighted) sends its query outward to every other token, receives weighted values in return, and produces an updated output. The bidirectional arrows indicate that every token attends to every other token simultaneously, and the crossing arrow beneath represents the weighted sum that combines those values into the final output. For each token, the transformer compares its query with the keys of all tokens. The resulting scores become weights, and those

weights are used to combine the corresponding values. This operation is usually written as [19]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{1.2}$$

Transformers usually repeat this operation in several heads and add positional information so the model knows where each token sits in the sequence. Multiple heads allow different relationships to be represented at the same time, while positional information preserves order [19].

1.6 Self-supervised learning

Machine learning methods can be grouped by how they receive feedback during training, as illustrated in Figure 1.9. Supervised learning trains on data with explicit labels: the correct answer is provided for every input, and the model learns to reproduce it. BC from Section 1.1.1 is a direct example: the expert demonstration acts as the label. Reinforcement learning instead receives feedback through a reward signal after taking actions in an environment; the model learns which state-action sequences lead to good outcomes. SSL sits between these two: it trains on unlabelled data by having the data itself provide the supervision signal. Part of the input is hidden, and the model learns to predict it from the remainder [21].

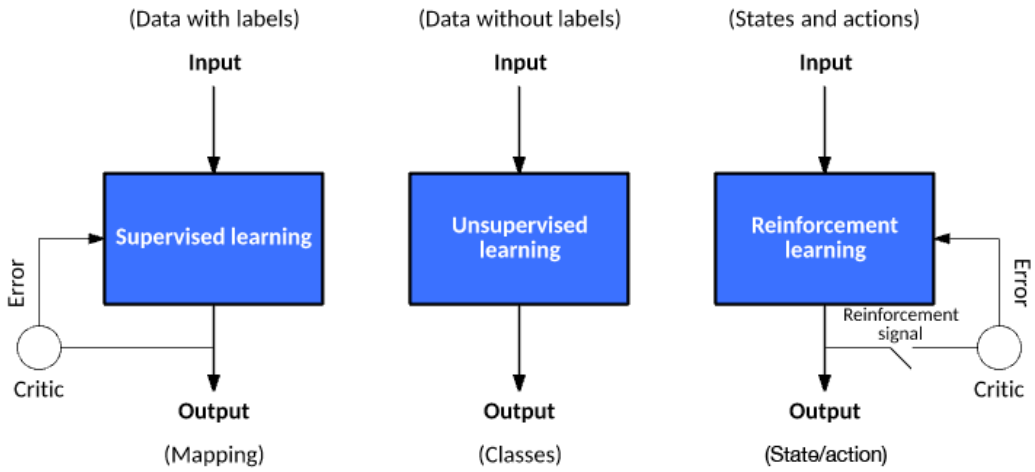


Figure 1.9 Three machine learning paradigms: supervised learning (explicit labels), reinforcement learning (reward signal), and self-supervised learning (unlabelled data provides its own supervision signal). The figure labels this third paradigm as unsupervised learning; the two terms are used interchangeably in this context [22].

Applied to video, SSL takes the following form: given a sequence of observed frames, hide one or more future frames and ask the model to predict them from the

frames that came before. No human annotation is required because the video recording itself provides the hidden target.

Predicting raw pixels runs into a problem described in Section 1.3: when the future is ambiguous the model cannot commit to a single outcome, so it predicts the average of all possibilities and produces a blurred image that does not match any real frame. Predicting in latent space avoids this. Instead of reconstructing every pixel, the model predicts the compact embedding of the future frame. Background motion and fine texture are discarded at the encoding step, so the predictor focuses on the structure that determines what actually happens next [14]. When the predictor is also given the action the agent intends to take, the result is a world model: a learned function that predicts how the scene embedding will change in response to a proposed action. The architectures that build on this idea are reviewed in Chapter 2.

2 Literature Review

This chapter surveys the literature motivating the three central design choices of this study's benchmark: that a specialist imitation policy is a strong base for bimanual manipulation, that a world model can usefully evaluate candidate action chunks in an offline benchmark, and that latent predictive learning is the right approach for building such an evaluator. Sections 2.1 and 2.2 review the two candidate imitation policies, ACT and Diffusion Policy, examining both the challenges they address and the trade-offs between them. Section 2.3 introduces world models as learned environment predictors, surveys the evaluator architectures most relevant to this role, and covers the three latent predictive architectures implemented in the benchmark, together with a negative control. Section 2.4 covers offline evaluation frameworks and their validity as proxies for real-world performance. The chapter closes with a summary of the literature gap that the benchmark addresses. The chapter assumes familiarity with the technical foundations in Chapter 1; world models are introduced here.

2.1 Action Chunking with Transformers (ACT)

ACT was introduced by Zhao et al. to address the compounding error and multimodal distribution challenges that make dexterous bimanual manipulation demanding for imitation learning [7]. On the A Low-cost Open-source Hardware System for Bimanual Teleoperation (ALOHA) bimanual platform it achieved success rates of up to 90% on precision tasks such as threading a cable tie, inserting a battery, and opening a translucent bag, outperforming behaviour cloning baselines and Generalist Agent (GATO) by substantial margins [7]. In SE(3) wrist-pose prediction specifically, small deviations in predicted position or orientation alter contact geometry in ways that cascade into grasp failure [1], and a policy trained with a naïve regression loss produces averaged predictions that may not represent any plausible trajectory when the demonstrated distribution is multimodal [8]. ACT's CVAE architecture addresses both of these failure modes directly.

ACT addresses both challenges through the use of a CVAE design (Section 1.4). The respective CVAE has an encoder-decoder structure depicted in Figure 2.1, where the encoder compresses an action sequence and joint observation into a style variable z that captures the multimodal distribution of plausible futures, and the decoder takes visual observations and joint positions, together with z, through a transformer encoder, and predicts a sequence of future action steps as an action chunk with a transformer decoder. The encoder is discarded at test time. In the original formulation, z is set to the mean of the prior (zero) to produce a single deterministic action chunk per

observation [7]. When a pool of candidate chunks is required, multiple values of z are sampled independently from the prior and each is decoded into a separate candidate, all conditioned on the same observation.

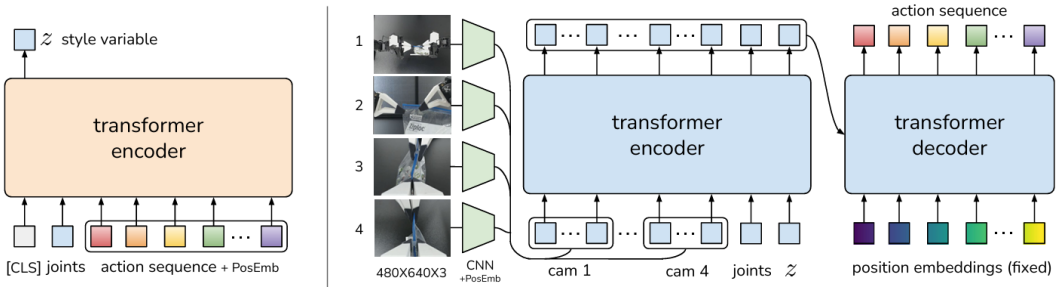


Figure 2.1 Architecture of the Action Chunking with Transformers (ACT) policy [7].

At inference time, the CVAE decoder produces each candidate action chunk by decoding an independently sampled z vector. A temporal ensembling technique is then applied: consecutive chunks overlap in their predicted time horizons, and a recency-weighted average is taken across overlapping predictions to produce the final action. This averaging smooths the executed motion and reduces jerkiness caused by abrupt chunk transitions. The result is a structured pool of plausible futures, all conditioned on the same observation, from which the ensembled output is drawn [7].

ACT carries known limitations alongside these strengths. The Gaussian prior on the style variable z constrains how faithfully the CVAE can represent highly multimodal action distributions: at transitions where qualitatively different trajectories are equally plausible, the latent space can average across modes rather than sample cleanly from each one, producing trajectories that are plausible in aggregate but imprecise at the most ambiguous decision points [8]. Chunk length is a fixed hyperparameter that must be tuned per task; longer chunks improve smoothness but commit the policy to a trajectory over a horizon that may no longer be appropriate as the scene evolves. Temporal ensembling introduces a recency-weighted lag that can slow the policy's response to sudden environmental changes [7]. These properties are well-documented in follow-on work comparing ACT with more expressive policy classes, and they motivate the inclusion of Diffusion Policy as the second candidate in the benchmark.

2.2 Diffusion Policy

Diffusion Policy offers a principled alternative approach to the same compounding error and multimodality challenges. Its core concept relies on starting from a noise distribution and iteratively refining action predictions through a diffusion process. By gradually denoising the predictions, Diffusion Policy captures complex multimodal

distributions effectively, producing plausible trajectories that align with the demonstrated data [8]. Figure 2.2 illustrates the policy overview:

- a) General formulation: at timestep t, the policy takes the latest T steps of observation data O_t as input and outputs T steps of actions A_t.
- b) CNN-based Diffusion Policy: FiLM (Feature-wise Linear Modulation) [23] conditioning of the observation feature O_t is applied to every convolution layer, channel-wise. Starting from K_t drawn from Gaussian noise, the output of noise-prediction network ϵ_θ is subtracted, repeating K times to get A_t^0, the denoised action sequence.
- c) Transformer-based Diffusion Policy: the embedding of observation O_t is passed into a multi-head cross-attention layer of each transformer decoder block. Each action embedding is constrained to only attend to itself and previous action embeddings (causal attention) using the attention mask illustrated [24, 25].

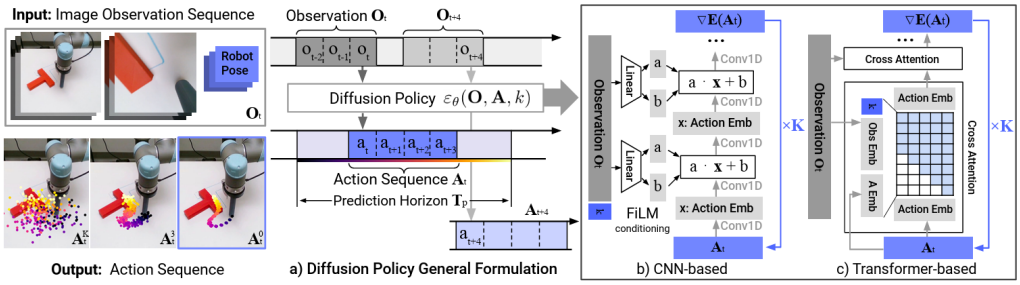


Figure 2.2 Diffusion Policy: general formulation (a), CNN variant (b), transformer variant (c) [8].

Chi et al. report that Diffusion Policy outperforms ACT on tasks with high action multimodality (such as multi-step object rearrangement and cup placement under visual ambiguity) while matching it on simpler reaching tasks [8]. The advantage arises because iterative denoising explores the full action distribution at each step rather than committing to a single latent mode as the CVAE prior must, making it more robust to the mode-averaging failure that affects ACT at ambiguous transitions.

The main limitation of Diffusion Policy is inference cost. Generating a single action chunk requires multiple sequential network evaluations: standard Denoising Diffusion Probabilistic Models (DDPM) variants use on the order of 100 denoising steps compared with the single forward pass of ACT's decoder, which substantially reduces the achievable control frequency [8]. Efficient transformer variants address this through progressive noise scheduling and parameter sharing, recovering much of the inference speed at added architectural complexity [25]. Both policies are included in the benchmark to span this representational trade-off: ACT's structured latent

sampling is computationally efficient but expressively constrained at high multimodality, while Diffusion Policy's iterative denoising is more expressive but more demanding at inference. Measuring the world model's effect across both allows the benchmark to determine whether evaluator gains are consistent across this axis.

2.3 World models

A world model is a learned predictor of how a system evolves over time. Given the current state and a proposed action, it estimates the state that is likely to follow. In a manipulation setting, the state may be a raw image, a latent representation of the scene (Section 1.3), or a structured description of hand and object configuration. Regardless of the representation chosen, the underlying principle is the same: the model internalises the dynamics of the environment and uses them to project forward from any given state [26].

World models become especially useful when a pool of candidate actions must be evaluated before any one is committed to. Each candidate is rolled forward through the model, producing a predicted future state; those predictions are ranked by predicted cost or consistency. The quality of this ranking depends heavily on the chosen representation: pixel-space world models must reconstruct every visual detail and suffer from mode-averaging at multimodal transitions, while models that operate in the compact learned latent space introduced in Section 1.3 can concentrate predictive capacity on the dynamics that matter for task outcomes [14]. The following subsections survey the literature supporting this evaluator design and describe the three world model architectures implemented in the benchmark.

2.3.1 World models as policy evaluators

WorldGym demonstrates that a learned world model used as a simulated environment for policy evaluation produces rankings that align with real-world task success with meaningful fidelity [27]. Candidate policies are evaluated not by deploying them physically but by rolling them out inside the model's predicted environment, reducing the cost of policy comparison to a forward pass. CLOVER extends this to closed-loop control, as shown in Figure 2.3. A visual planner generates a sequence of sub-goal frames; the executor measures the embedding-space error between the predicted sub-goal and the current observation. When the error exceeds a threshold, the system triggers an adaptive transition and replans rather than committing to the original trajectory. The feedback loop, absent from open-loop imitation policies, is the mechanism that translates world model prediction accuracy into task completion

gain [28]. Together, WorldGym and CLOVER establish that world model rankings are informative both before and during execution.

The broader question of whether such evaluator rankings transfer to physical outcomes, and the qualifications around out-of-distribution reliability, are examined in Section 2.4.

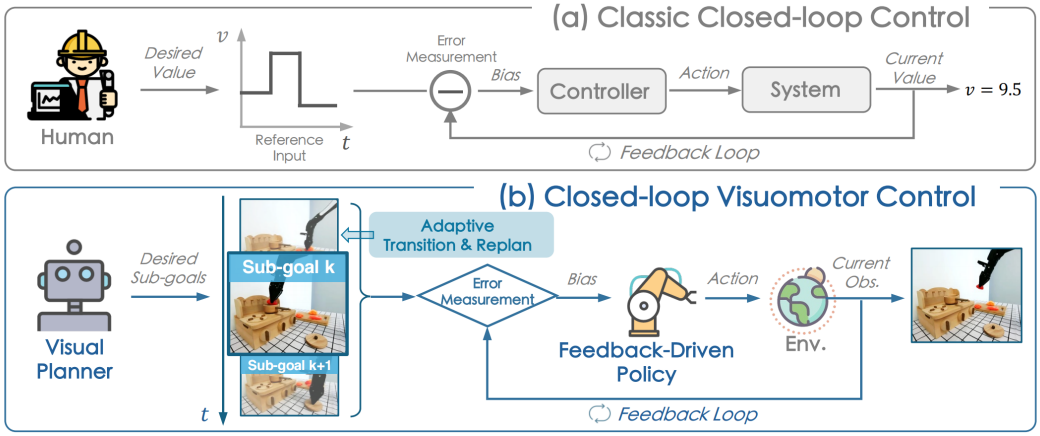


Figure 2.3 Classic closed-loop control (a) versus CLOVER's closed-loop visuomotor control (b) [28].

The practical value of this evaluator loop is measurable. Table 2.1 reproduces CLOVER's real-world evaluation on a multi-step robot manipulation suite, reporting the average number of consecutive sub-tasks completed (`avg_len`) for representative baselines and CLOVER [28].

Table 2.1 Average consecutive sub-task completions (`avg_len`) on a real-world multi-step manipulation suite. Higher is better. CLOVER's closed-loop feedback via world model prediction nearly doubles the completion depth of the strongest open-loop baseline. Data from [28].

Method	avg_len
ACT [7]	0.6
R3M (pretrained visual repr.) [29]	0.7
RT-1 [30]	1.1
CLOVER (closed-loop)	**2.1**

Alongside the task completion gain, CLOVER reduces embedding-space prediction Root Mean Square Error (RMSE) relative to open-loop baselines by correcting the policy's course when the world model detects a divergence from the expected trajectory, demonstrating that prediction accuracy and task success move together under the evaluator framing.

2.3.2 Latent predictive architectures

The energy-based framework of LeCun and Dawid establishes why prediction should operate in latent rather than pixel space [14]: the same representational argument introduced in Section 1.3 applies directly to the prediction target, favouring compact embeddings over pixel reconstructions that force the model to account for irrelevant visual detail and average over all plausible continuations at ambiguous transitions.

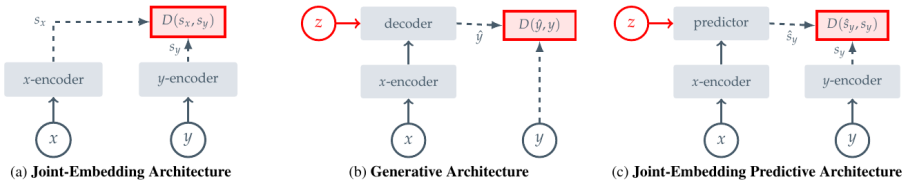


Figure 2.4 Three SSL prediction paradigms: joint-embedding (a), generative (b), and joint-embedding predictive (c) [21].

Image Joint Embedding Predictive Architecture (I-JEPA) establishes the joint-embedding predictive architecture as a viable approach to learning rich visual representations without any pixel reconstruction objective [21]. Panel (c) of Figure 2.4 shows the data flow directly: a context block x from the image is passed through the context encoder to produce latent tokens z_x. A lightweight ViT predictor then takes z_x together with positional tokens indicating where the masked target regions y are located, and outputs predicted latent representations $\hat{s}_y$ for those regions. A separate target encoder, whose weights are a slow exponential moving average of the context encoder, independently encodes the actual target pixels into ground-truth latents s_y. The loss $\text{MSE}(\hat{s}_y, s_y)$ is computed entirely in latent space: no decoder, no pixel generation, no reconstruction of lighting or texture. The outcome is a context encoder whose representations capture semantically meaningful structure (object shape, spatial layout, and scene dynamics) because that is all the predictor ever needs to get the target embeddings right. Training is approximately 2.5 times faster than pixel-reconstruction methods as a result. V-JEPA2 extends this same joint-embedding predictive architecture from images to video sequences.

Four architectures are covered in the following subsections, each representing a distinct representational strategy. V-JEPA2 and V-JEPA2-AC share the same pretrained video backbone: V-JEPA2 is the internet-scale pretrained encoder, while V-JEPA2-AC extends it with an action-conditioned transition head and is the form used directly in the benchmark. DexWM uses a pretrained image backbone with specialised hand-geometry supervision. LeWorldModel (LeWM) learns all representations from scratch on the task data itself.

V-JEPA2

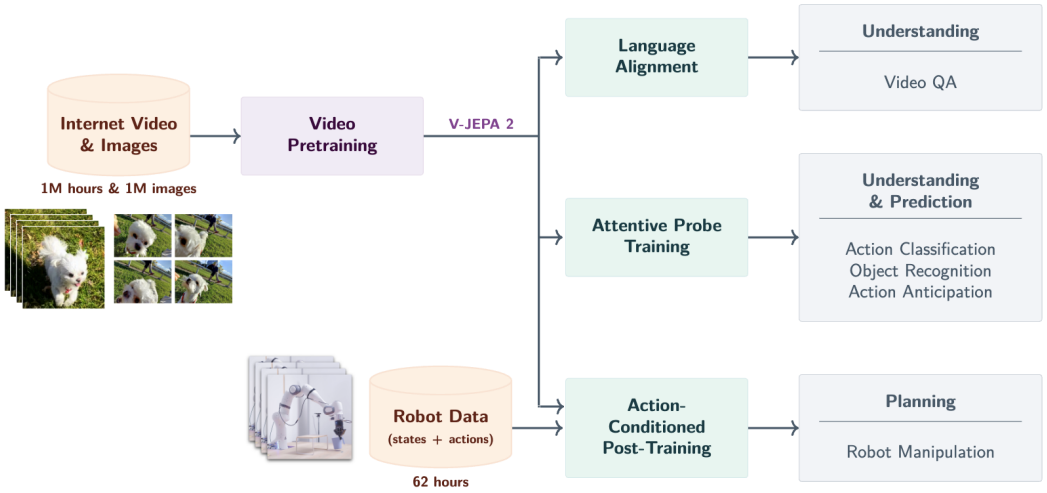


Figure 2.5 V-JEPA2 post-training pathways from internet-scale backbone to specialised applications [31].

V-JEPA2 extends the I-JEPA framework from static images to video, pretraining on over one million hours of internet video [31]. The pretraining shifts from spatial to spatio-temporal masking: the context encoder receives past observation frames, and the predictor must recover future latent states. The training objective combines teacher-forcing with multi-step autoregressive rollout losses, encouraging accurate short-horizon predictions and consistent long-horizon trajectory representations. The result is a frozen ViT-Large encoder, approximately 300 million parameters, that encodes physical priors about hand and object interactions acquired from internet-scale pretraining. As shown in Figure 2.5, the pretrained backbone supports three post-training pathways: Language Alignment, which grounds visual embeddings in natural language; Attentive Probe, which adds a lightweight classification head; and Action Conditioning, which extends the predictor to take robot actions as input and forms the basis of V-JEPA2-AC.

V-JEPA2-AC

The frozen ViT-Large encoder from V-JEPA2 serves as the backbone of V-JEPA2-AC, producing a fixed latent embedding for each observation frame with no gradient updates to the backbone weights. A lightweight action-conditioned transition head is trained on top of these stored embeddings: it takes the current frame latent together with a proposed action chunk (robot joint states and end-effector poses) as the conditioning signal z, and predicts the latent representation of the future frame. Candidate action chunks are then scored by comparing the predicted future embedding against the actual observed future embedding under an L1 objective. This

design, shown on the right of Figure 2.6, separates what the network must learn into two parts: the frozen encoder supplies rich visual features that generalise from internet-scale pretraining, and the small trained head learns only how actions propagate through those features.

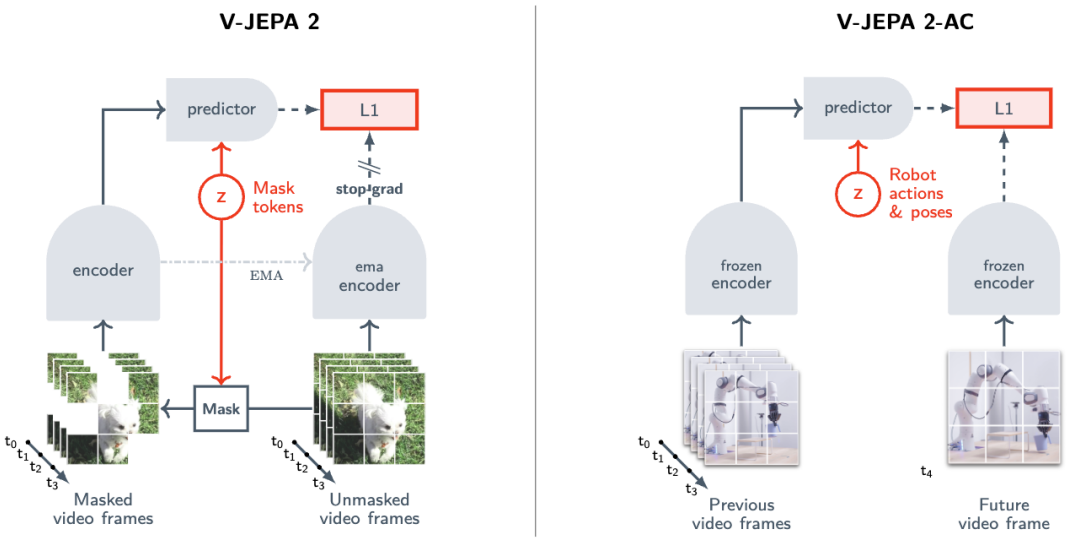


Figure 2.6 Standard V-JEPA2 (left) versus action-conditioned V-JEPA2-AC (right) [31].

An important qualification is raised by Tsagkas et al., who show that pretrained visual representations can encode spatial plausibility well while providing a weaker signal for fine-grained temporal dynamics [32]. A frozen encoder trained on internet video may represent hand shape and position accurately at each individual frame while providing a less reliable signal about whether the predicted trajectory is kinematically consistent across the full chunk horizon. This temporal limitation provides context for the path shape results in Chapter **??**.

DexWM

DexWM [33] is a dexterous manipulation world model from Meta FAIR. As shown in Figure 2.7, its pipeline encodes each input frame with a frozen DINOv2 image encoder, producing a grid of spatial patch tokens rather than a single global embedding. These patch-level representations preserve fine-grained spatial information about hand positions and object locations, which gives the predictor strong generalisation across diverse environments and viewpoints. A CDiT predictor then takes the spatial latent tokens together with action and camera-motion inputs to predict the next latent state, and a lightweight decoder maps the result back to image and keypoint space. Where V-JEPA2-AC uses a video-pretrained backbone, DexWM uses a backbone pretrained on static images; the comparison therefore isolates the contribution of video-specific temporal pretraining to world model quality.

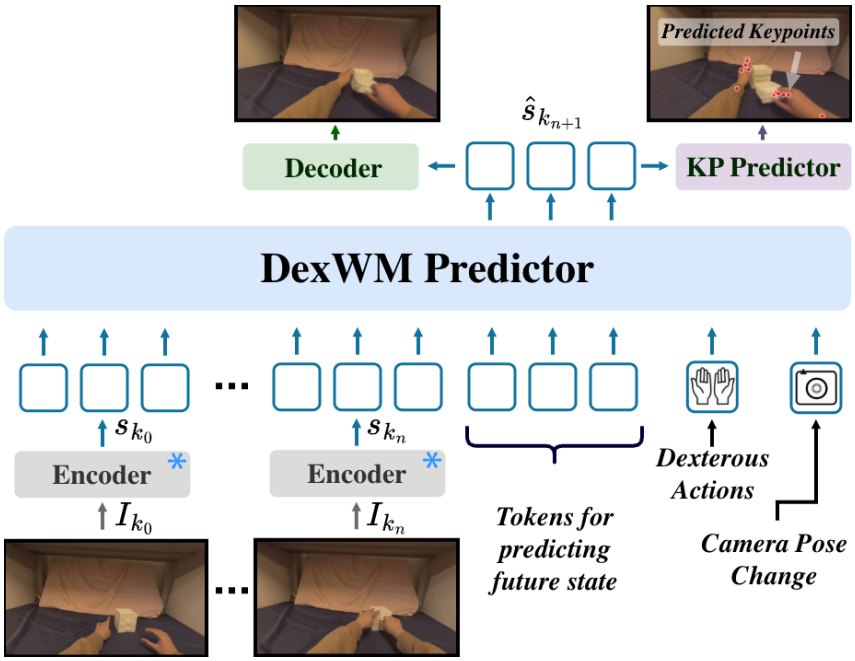


Figure 2.7 High-level layout of the DexWM world model pipeline, illustrating how predicted keypoints are incorporated alongside the encoder, CDiT predictor, and decoder stages [33].

A distinguishing feature of DexWM is the Hand Consistency (HC) loss: an auxiliary transformer head predicts fingertip and wrist heatmaps from the latent state, imposing a geometric prior that the predictor's latent trajectory must remain consistent with plausible hand configurations. This explicit hand-geometry supervision is absent from both V-JEPA2-AC and LeWM, whose latent spaces are shaped only by prediction error or general regularisation. Goswami et al. report a 50% improvement over Diffusion Policy on grasping and placing tasks, with an 83% zero-shot real-world grasping success rate under latent planning [33].

LeWorldModel

LeWM is a latent world model trained end-to-end from random initialisation on raw pixel observations [34]. Its architecture consists of two components: an encoder that maps each frame o_t to a compact latent vector z_t, and a predictor that models environment dynamics by estimating the next latent embedding $\hat{z}_{t+1}$ from z_t and the action a_t. The encoder is a ViT-Tiny with approximately 15 million parameters. Each frame is represented as a single 192-dimensional token rather than a spatial patch grid, which keeps the latent space compact and makes planning significantly faster than models that produce higher-dimensional patch representations.

As shown in Figure 2.8, the training pipeline consists of an encoder that maps each observation frame to a compact latent vector, a dynamics predictor that autoregressively estimates the next latent given the current embedding and action, and

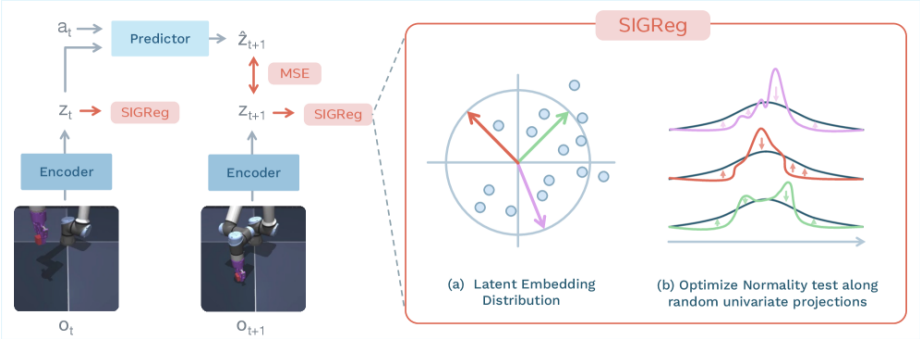


Figure 2.8 LeWM training pipeline: encoder, dynamics predictor, and SIGReg regulariser [34].

a SIGReg regulariser that enforces a Gaussian prior on the latent distribution by optimising normality along random univariate projections. The training objective reduces to two terms:

$$\mathcal{L}_{\text{LeWM}} = \mathcal{L}_{\text{pred}} + \lambda\,\text{SIGReg}(Z) \tag{2.1}$$

where $\mathcal{L}_{\text{pred}}$ penalises inaccurate next-embedding forecasts and $\text{SIGReg}(Z)$ stabilises end-to-end training without contrastive negative pairs or an exponential moving average target encoder [34]. This reduces the number of tunable loss hyperparameters from six, as required by prior Joint Embedding Predictive Architecture (JEPA)-style designs, to one.

DINO-WM, the closest comparable baseline, follows a similar latent prediction objective but uses a frozen DINOv2 backbone producing a grid of spatial patch tokens per frame [35]. Like DexWM, DINO-WM therefore carries the spatial resolution and generalisation benefits of DINOv2's patch-level representations, but at substantially higher computational cost. LeWM replaces the frozen backbone with a fully learned ViT-Tiny encoder that collapses each frame to a single 192-dimensional token, trading spatial resolution for training speed. In practice, LeWM can be trained on a single GPU in a few hours and achieves a 48$\times$ planning speedup over DINO-WM [34]. DexWM's CDiT predictor operating on full patch grids is significantly slower to train and requires substantially more GPU memory; this difference in training cost is expected to widen further at larger dataset scales and is quantified in Chapter 4. Despite its compact size, LeWM's latent space encodes physically meaningful structure: the model detects physically implausible state transitions and captures spatial information such as agent and object locations across frames.

The three-way comparison across V-JEPA2-AC, DexWM, and LeWM is therefore not simply an architecture comparison. It is a comparison between fundamentally different representational strategies: video-pretrained transfer (V-JEPA2-AC), image-pretrained transfer with geometric supervision (DexWM), and

self-supervised task-data learning from scratch (LeWM). This distinction makes the comparison principled and is the basis for the experimental design in Chapter 3.

PointWorld

PointWorld is a large pretrained 3D world model for in-the-wild rigid arm manipulation [36]. Unlike the latent-space models described above, it operates directly in 3D geometry: scene state is lifted from RGB-D images into a full-scene point cloud, and robot actions are represented as the 3D point flows traced by robot surface geometry through forward kinematics. This embodiment-agnostic representation allows the model to reason over physical interaction geometry rather than learned abstract features.

Such a model is not designed to handle a 16-dimensional dexterous hand action space, where fine-grained finger articulation drives the manipulation outcome. The forward-kinematics representation assumes rigid arm geometry, making the model structurally mismatched when applied to dexterous bimanual hand motion.

2.4 Offline evaluation

A central concern for any offline benchmark is whether its metrics produce rankings that are meaningful proxies for real-world performance. Without this property, an offline evaluation framework reduces to a measure of in-distribution accuracy with no predictive validity for deployment.

At large scale, DreamerV3 demonstrates the effectiveness of latent-space candidate evaluation across diverse continuous control domains, including locomotion and manipulation [26]. DreamerV3 is an online reinforcement learning system, not an offline benchmark; it is included here because it provides large-scale empirical evidence that latent-space evaluation is an effective selection mechanism when properly calibrated, which motivates its use in the offline setting examined in this study. Its world model maintains a compact recurrent latent state that transitions under action conditioning; candidate trajectories are rolled out entirely inside this imagined space and scored by a learned value function, with no return to the real environment between evaluations. TD-MPC2 confirms the principle with a scalable planner that optimises a temporal-difference value function defined over predicted latent trajectories rather than observed states [37]. Both systems show that ranking candidate actions by predicted latent-space outcomes is competitive with direct online evaluation across a range of motor control benchmarks.

A critical qualification is that latent-space scoring becomes unreliable outside the training distribution. MOPO, MOReL, MBOP, and MOPP each establish that

learned models produce inaccurate predictions for out-of-distribution states, requiring conservative candidate selection to avoid trajectory degradation [38–41]. None of these works addresses dexterous manipulation specifically; they benchmark locomotion and simple control tasks, leaving open how well the principle transfers to the fine-grained wrist-pose prediction setting.

Two further bodies of work establish that offline rankings can transfer to physical outcomes when the evaluation domain is well-calibrated.

SimplerEnv establishes the correspondence between simulation-based evaluation and physical robot performance for visuomotor policies [42]. The study evaluates multiple manipulation tasks and policy types in simulation and then physically, using a WidowX robot under matched visual and control conditions. Rankings produced from simulation-only evaluation closely track rankings from physical trials, with correlations holding across both policy families and task types. The key condition for transfer is domain calibration: visual appearance, camera viewpoint, and control interface must match between the simulation and the physical system. When these are well-matched, simulation rankings provide a reliable ordering of policies without requiring physical deployment.

WorldEval extends the validation to the world-model setting [43]. Rather than evaluating policies in a manually constructed simulator, WorldEval scores them using a learned world model that predicts future states from offline trajectory data. The world-model-based rankings correlate strongly with real-world task success: policies that score well in world-model evaluation also tend to succeed when deployed physically.

Both results carry a shared qualification: rankings hold under well-calibrated conditions, where the evaluation domain is consistent with the training distribution and the deployment context. When this alignment breaks down, offline rankings can degrade. The design implications of this constraint are discussed in Chapter 3.

Summary. The literature surveyed in this chapter establishes three supporting findings. First, world model rankings are informative evaluators both before and during execution, as demonstrated by WorldGym and CLOVER [27, 28]. Second, latent predictive objectives produce more efficient and better-calibrated representations than pixel-level reconstruction, as demonstrated by the JEPA family [21, 31]. Third, offline evaluation rankings can carry predictive validity for physical outcomes when the evaluation domain is well-calibrated, as demonstrated by SimplerEnv and WorldEval [42, 43].

These findings leave two questions unanswered, which this dissertation addresses directly.

RQ1. Does augmenting a fixed imitation-learning policy with a latent world model

evaluator improve candidate selection quality in an offline dexterous manipulation benchmark, and does any improvement hold across policy families?

The offline reinforcement learning literature establishes the need for principled candidate selection but benchmarks only locomotion and simple control tasks [38–41]. No controlled study isolates the world model evaluator's contribution on a fixed dexterous manipulation policy.

RQ2. Does the world model's representational strategy (pretrained video backbone, task-data-trained from scratch, or out-of-domain modality) produce meaningfully different selection guidance when applied to the same policy on the same evaluation data?

The interaction between world model architecture and policy performance has not been examined in this setting. Answering RQ2 requires a benchmark that holds the policy and evaluation surface fixed while varying only the world model family. The methodology for constructing and running such a benchmark is described in Chapter 3.

3 Methodology

This chapter describes the experimental design used to evaluate whether latent world model augmentation improves candidate selection quality when paired with imitation-learning policies, and whether any observed effect generalises across policy families and world model architectures. Five design decisions structure the evaluation: the choice of dataset and action space (Section 3.1); a three-branch framework that isolates the contribution of each selection mechanism (Section 3.2); the selection of two policy families that span distinct generation mechanisms (Section 3.3); the selection of five world model families that cover a range of pretraining regimes and input modalities (Section 3.4); and a fair experimental protocol that normalises training exposure across all trained components (Section 3.5). Evaluation metrics and scope limitations are addressed in Sections 3.6 and 3.7.

3.1 Dataset and Experimental Setup

3.1.1 EgoDex: source and selection rationale

The benchmark is built on EgoDex [44], a large-scale egocentric dataset of bimanual dexterous manipulation collected using Meta Quest Pro head-mounted cameras. It spans more than 100 task categories across its full release, including folding, slotting, pouring, and kneading, covering a wide range of fine-grained hand motions. The full release comprises five training parts and supplementary additional-data splits; this study uses the first three training parts, which together comprise more than 194,000 demonstration episodes and approximately 1.3 TB of paired video and annotation data. Each episode provides synchronised RGB video and full six-degree-of-freedom $\mathrm{SE}(3)$ wrist annotations for both hands at approximately 30 frames per second.

The choice of dexterous manipulation as the evaluation domain is not incidental. This dissertation forms part of the University of Malta's ongoing M.AI.ESTRO research project [45], which investigates the use of AI for scene and task prediction on dexterous manipulation tasks to support industrial transfer learning and human skill digitisation. The research domain was therefore set by this institutional context: the methodology operates on human hand demonstrations rather than robotic end-effector data because the target application is the transfer of tacit human dexterous skill. An evaluation framework built around robot arm trajectories would not address that objective. The introduction situates this motivation more broadly; the present chapter focuses on the methodological consequences of the domain choice.

Within the dexterous manipulation setting, EgoDex was selected on three

properties. First, the benchmark requires a dexterous bimanual hand action space: fine-grained articulation across an eighteen-dimensional bimanual wrist pose representation drives the manipulation outcome, and large-scale corpora built around rigid gripper or arm contact do not provide this. Second, the egocentric viewpoint aligns with the intended downstream deployment setting of a wearable assistive device. Third, the dataset's scale and task diversity are sufficient to support simultaneous training of five world model families and two policy architectures on declared non-overlapping splits while retaining a held-out test set of meaningful size.

EgoDex carries two acknowledged limitations. The demonstrations are produced by human subjects rather than a robotic system, which introduces a kinematic domain gap between the training distribution and any robotic deployment. No contact or force signals are present; the action representation is purely kinematic $\mathrm{SE}(3)$. Both are accepted as known constraints rather than experimental variables and are discussed further in Section 3.7.

3.1.2 Windowing protocol

The dataset is stored as paired `.mp4` and `.hdf5` episode files. A deterministic index stage discovers all valid paired episodes, infers their lengths from HDF5 metadata, and writes canonical observation and prediction windows for each split. This shared index is the single source of truth for all downstream modules: policy training, world model extraction, and evaluation all consume the same window file for their respective split, ensuring that output differences across the experimental branches can be attributed to branch design rather than to differences in the data presented to each component. The benchmark uses a fixed-stride windowing scheme as the canonical contract; a supplementary state-change-targeted scheme was developed during early exploration and used exclusively for qualitative inspection.

Strategy 1: Fixed-stride sampling. Each window spans 16 observation frames followed by 16 prediction frames, with consecutive window starts separated by a stride of 128 frames. At 30 frames per second this places starts approximately 4.3 seconds apart, corresponding to an observation context and prediction horizon of approximately 0.53 seconds each. This contract is encoded as `obs16_pred16_s128` and forms the foundation for all policy training and world model training (see Figure 3.1).

Strategy 2: State-change-targeted sampling. The fixed-stride scheme samples the episode timeline uniformly, which causes it to under-sample high-motion events: long near-static segments contribute windows at the same rate as active manipulation periods. As detailed in Chapter 4, this limitation motivated a revised strategy adopted for a subsidiary perceptual inspection task and qualitative development review. Rather

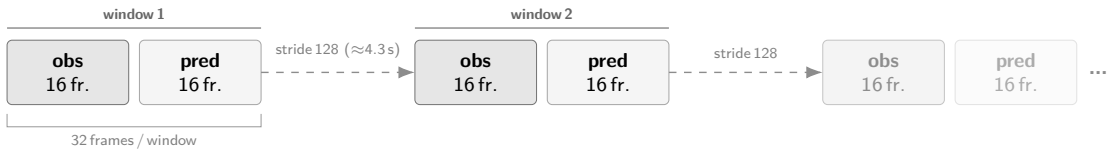


Figure 3.1 Fixed-stride windowing scheme (`obs16_pred16_s128`) used as the canonical training and evaluation index. Author-generated schematic.

than placing windows at fixed intervals, this approach computes smoothed hand-state change scores from the wrist transform and confidence sequences, selects candidate windows near high-change peaks, and applies episode-first task-stratified sampling with a fixed random seed (see Figure 3.2). A relative-position fallback retains slower episodes that would otherwise be excluded, yielding one frozen window per episode stratified across all represented task categories.

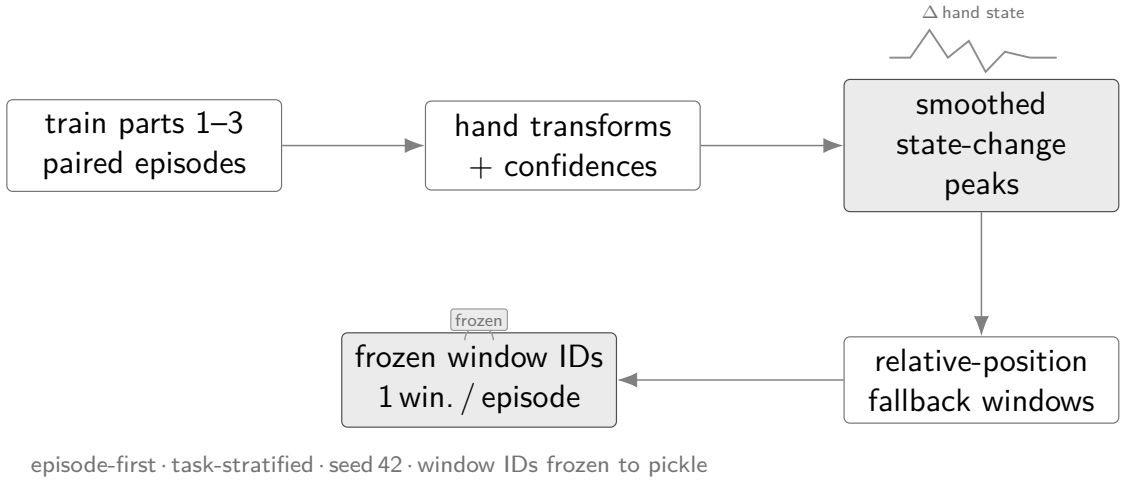


Figure 3.2 State-change-targeted sampling pipeline yielding one frozen window per episode. Author-generated schematic.

The canonical training and evaluation index uses the fixed-stride scheme across all experimental branches. The state-change-targeted subset was used exclusively for a subsidiary perceptual inspection task and qualitative development review; it does not form part of the primary benchmark evaluation.

> **Writing note.** All benchmark results reported in this thesis use the fixed-stride contract (Strategy 1) exclusively. Strategy 2 has been implemented for qualitative development inspection. A systematic Strategy 1 vs. Strategy 2 comparison is not carried out within this thesis but represents a natural future direction. If a full Strategy 2 evaluation were completed, the following aspects of this chapter would require revision:
>
> - *This section:* the framing of Strategy 2 as a qualitative inspection tool would be updated to describe it as a parallel primary evaluation contract with its own training and evaluation index.

- *Section 3.1.3:* the training-splits table would be extended to include Strategy 2 window counts per split and task, and the test index description updated accordingly.
- *Section 3.6:* result tables and figures would require explicit strategy labels, and any cross-strategy comparison would warrant a dedicated analysis subsection.
- *Chapter* **??***:* all results would need split-by-strategy labelling to distinguish which sampling contract each row corresponds to.

3.1.3 Training splits and test set

Three cumulative training splits were assembled from the first three parts of the EgoDex release, each a strict superset of the previous; Table 3.1 summarises their composition. EgoDex ships five training parts plus supplementary additional-data splits; parts four and five and all additional-data splits are excluded from this study to bound the experimental scope. The held-out test split is the official EgoDex test set and is never used during training at any stage by any model or policy. For the Strategy 1 benchmark [Possible Reword with Strategy 2], the same fixed-stride test index is used for quantitative evaluation and for the qualitative review examples, keeping the reported comparisons aligned with the training contract in Table 3.1.

Table 3.1 EgoDex splits under the `obs16_pred16_s128` windowing contract. Episode and window counts are derived from the canonical index.

Split	Tasks	Episodes	Windows	Role
`train_part1`	26	46,091	140,239	Stage 1 training
`train_part1_2`	39	140,654	291,369	Stage 1+2 training
`train_part1_2_3`	63	194,333	430,962	**Canonical training**
`test`	111	3,229	7,607	**Frozen evaluation**

The episode counts reflect the uneven collection density across parts: the 13 task categories added in part two average approximately 7,300 episodes each, roughly four times the average of the 26 part-one tasks, because part two focuses on high-frequency foundational primitives such as pick-and-place and folding that were collected at greater volume.

The canonical results in Chapter **??** use `train_part1_2_3` across all trained components. The test split is the official EgoDex held-out set spanning 111 task categories, which includes all 63 categories present in `train_part1_2_3` as well as 48 categories drawn from the excluded parts; evaluating on this broader set provides a measure of out-of-distribution generalisation. Its window index is frozen; no post-hoc filtering or re-sampling is applied. The incremental training curriculum across the three

stages is described in Chapter 4.

3.2 Three-Branch Experimental Framework

The central question of this study is whether a latent world model improves candidate selection quality when paired with an imitation-learning policy, and whether that effect generalises across policy families. Answering both questions cleanly requires holding every other experimental factor constant. The benchmark therefore organises evaluation around three branches that share the same policy weights and the same evaluation windows; the candidate pool composition and selection rule differ across branches (Figure 3.3). Any measured performance difference between branches is intended to reflect the selection mechanism, since training data, architecture, and window coverage are held constant across branches.

Branch 1 (B1): Policy-only baseline. The policy generates its full candidate pool and selects by its own internal preference; no world model is queried. The ideal design produces several structurally independent candidates so the downstream branches can compare meaningfully different trajectories. B1 establishes the trajectory prediction quality each policy achieves acting alone and is the baseline against which B2 and B3 are measured.

Branch 2 (B2): World-model-inclusive selection. The world model scores all candidates in the pool and selects the highest-scoring one unconditionally; no margin threshold is applied. The candidate pool is the same as B1. Scoring proceeds by predicting forward in the world model's latent space from the current observation and measuring prediction fidelity against each candidate action. The policy's own output is not privileged; it is selected only if the world model assigns it the highest score. For V-JEPA2 without action conditioning, the transition head required to produce candidate-specific latent predictions is absent; if the model assigns indistinguishable scores to all candidates, B2 degenerates to B1, directly testing whether action conditioning is the operative mechanism.

Branch 3 (B3): World-model-exclusive selection. The policy's preferred output is excluded from the evaluation pool; the world model selects the highest-scoring remaining candidate unconditionally. The ideal design ensures the remaining candidates are structurally independent so that exclusion meaningfully changes the selection outcome. B3 tests whether the world model maintains selection quality when the policy's top choice is unavailable. All five world model families apply this protocol.

Two policy families are evaluated: ACT and Diffusion Policy (DP) (Figure 3.3). For DP the three-branch progression follows the ideal description above. Five independent reverse-diffusion chains seeded from distinct noise samples serve as

candidates. B1 uses DP's policy-internal argmax over those five; B2 passes the same pool to the world model scorer; B3 excludes DP's preferred sample and scores the remaining four independent chains.

ACT presents a different case. CVAE posterior collapse in the trained checkpoint means ACT produces one deterministic output per window ($N{=}1$); investigation and three retraining attempts are documented in Section 4.4. A CEMP planner generates four additional candidates for B2 and B3 by sampling SE(3) perturbations around ACT's output. The branch pools therefore differ from the ideal: B1 contains ACT's single output ($N{=}1$); B2 contains that output plus four CEMP-planned alternatives ($N{=}5$); B3 contains the four CEMP perturbations only. Because all four candidates are anchored to ACT's prediction, B3 for ACT is a world-model-guided local search rather than selection from a policy-independent pool. Candidate generation details are in Section 3.3.

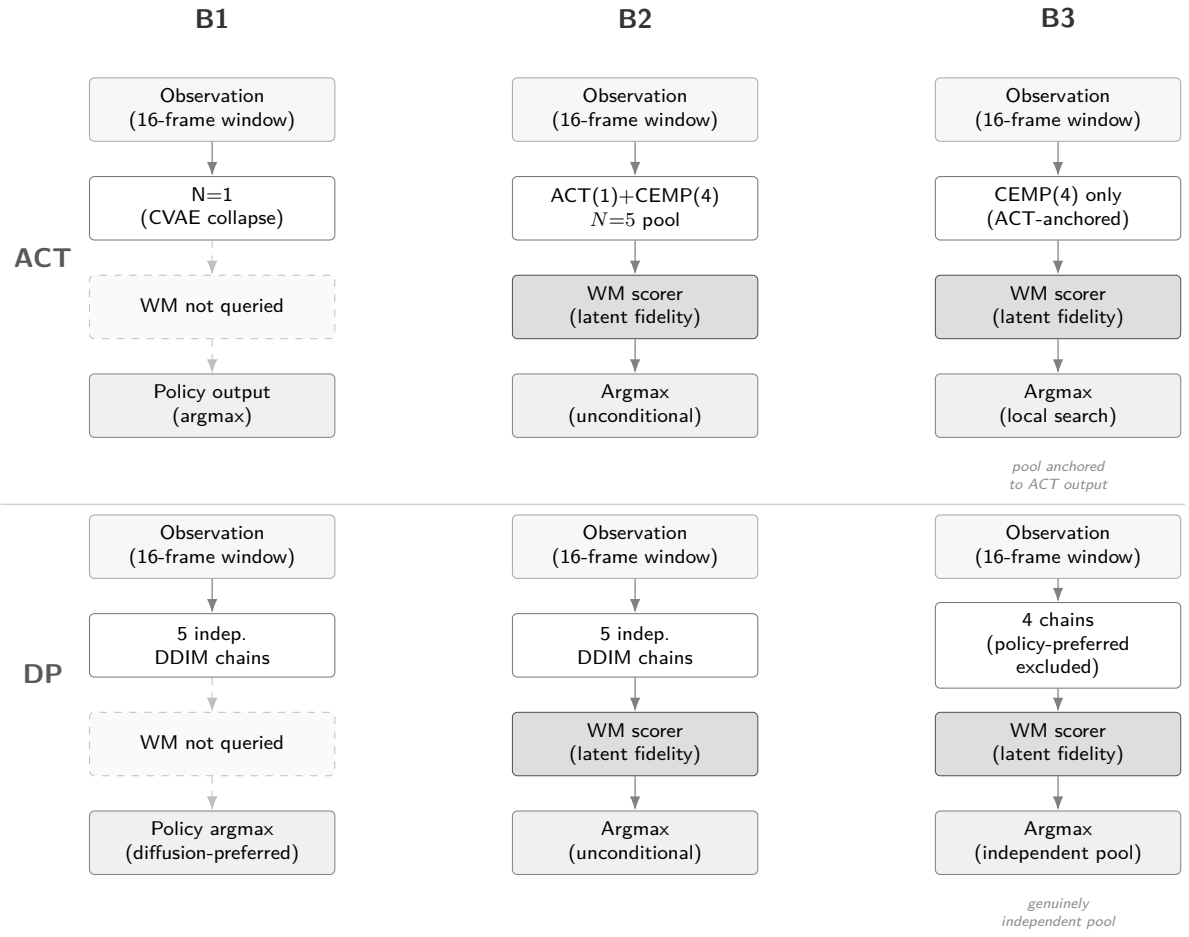


Figure 3.3 Three-branch evaluation framework (Section 3.2). ACT row uses CEMP candidates due to CVAE collapse; DP row follows the ideal design.

3.3 Policy Variants

The benchmark evaluates two imitation-learning policy families. Their architectures and training objectives are reviewed in Chapter 2; the properties relevant to this study are how each generates its candidate pool and what that implies for world-model scoring.

- **Action Chunking Transformer** (Section 2.1). ACT is architecturally designed to generate diverse candidates by sampling latent vectors $z \sim \mathcal{N}(0, I)$ from its CVAE prior and decoding each into a 16-step bimanual SE(3) action chunk. In the trained checkpoint used here, CVAE posterior collapse is observed: the decoder is insensitive to z at inference time, producing identical outputs regardless of the sampled vector (maximum output deviation under $z{=}10$ is $3{\times}10^{-7}$). ACT therefore contributes one deterministic candidate per window ($N{=}1$). The single forward pass requires one network evaluation. Three retraining attempts to resolve the collapse and the subsequent adoption of CEMP for B2/B3 candidate diversity are documented in Section 4.4.
- **Diffusion Policy** (Section 2.2). DP generates each candidate through an independent reverse-diffusion chain using a Denoising Diffusion Implicit Models (DDIM) schedule of ten steps (Section 2.2) [46]. Producing five candidates requires five separate chains and approximately fifty network evaluations per window, roughly an order of magnitude more than ACT. The Diffusion Transformer (DiT) backbone is used. Candidates are structurally independent because each starts from a fresh noise sample; diversity arises from stochastic initialisation rather than from a shared latent manifold.

Evaluating both families under the same three-branch protocol allows the study to ask whether world-model augmentation improves candidate selection in general or only under a particular generation mechanism. A result that holds across both constitutes stronger evidence for the generality of the approach.

3.4 World Model Families

Five world model families are evaluated as candidate selectors within the three-branch framework (Section 3.2). Their architectures and pretraining objectives are reviewed in Chapter 2; this section describes each family's methodological role. The five families span a deliberate range: from a large pretrained video backbone through lightweight from-scratch models to an action-specificity negative control, providing multiple axes of comparison.

At evaluation time each world model receives the current observation and the candidate action chunks from the pool, predicts forward in its latent space, and returns a scalar quality measure per candidate. Policy weights and evaluation windows are fixed; the only variable is which world model scores the pool.

V-JEPA2-AC: Pretrained video backbone. The architecture is reviewed in Section 2.3.2. A frozen V-JEPA2 [31] Vision Transformer (ViT)-Large backbone provides a 1024-dimensional frame embedding pretrained on large-scale video via the JEPA objective. A lightweight action-conditioned transition head, trained on benchmark data, maps a current embedding and an action chunk to a predicted future embedding; the L2 distance between prediction and observed future embedding serves as the candidate score. This family represents the upper end of the representational hierarchy: rich, general-purpose video features acquired at scale before any manipulation-specific training.

LeWM: From-scratch latent world model. The architecture is reviewed in Section 2.3.2. LeWM [34] is trained entirely from raw EgoDex pixels with no pretrained weights. A ViT-Tiny encoder and decoder are jointly optimised via a next-embedding prediction loss and a Gaussian regularisation term, without exponential moving average stabilisation or reconstruction of pixel values. This family represents the opposite end of the pretraining spectrum: all representations are learned from the benchmark training data alone, making it a provenance-clean baseline with no external data dependency. The expectation is that task-domain training from scratch compensates for the absence of video pretraining, testing whether manipulation-domain supervision alone is sufficient for action discrimination at this scale.

DexWM: Domain-specific architecture. The architecture is reviewed in Section 2.3.2. DexWM [33] pairs a frozen DINOv2 image encoder with a CDiT predictor and incorporates a hand-consistency loss that explicitly supervises fingertip and wrist heatmap predictions. It is included to assess whether a model built with explicit hand-geometric supervision outperforms general video predictors on this task. A constraint on the training setup applies: as described in Section 3.7, publicly available model weights could not be sourced and the model was therefore trained from scratch on the benchmark's declared training splits, adopting the same provenance stance as LeWM: no external pretraining and no data outside the declared splits. Despite this constraint, the domain-specific inductive bias of the hand-consistency loss provides a comparison point for whether explicit hand-geometry supervision compensates for the absence of video pretraining.

V-JEPA2 (no action conditioning): Ablation control. This family uses the same frozen V-JEPA2 backbone as V-JEPA2-AC but omits the action-conditioned transition head. The proposed action chunk is not passed to the model. The future embedding is

predicted from visual context alone via a learned visual-only prediction head (Section 4.5.4); the candidate score therefore reflects visual plausibility rather than action-conditional consistency. The purpose is to isolate the contribution of action conditioning: if V-JEPA2-AC outperforms this variant, the advantage is attributable to the model's ability to reason about action consequences rather than to the representational power of the backbone alone. The expected outcome for this variant is a null result at B2 (zero win rate against B1), because action-blind scores are indistinguishable across candidates regardless of what action each proposes.

PointWorld: Action-specificity negative control. This family uses the same frozen V-JEPA2 backbone and transition head as V-JEPA2-AC, but the candidate action chunks are degraded before scoring. Each 18-dimensional bimanual $\mathrm{SE}(3)$ chunk is projected to a 7-dimensional robot arm space by extracting wrist orientation from the left hand only and collapsing finger motion to a scalar aperture, then zero-padded back to 18 dimensions. The right hand is discarded entirely; the 11 recovered dimensions carry no meaningful hand-motion information. The transition head therefore receives the correct visual features but systematically wrong action conditioning. This tests a different hypothesis from the V-JEPA2-No-AC ablation: whereas that variant passes zero actions (no conditioning at all), PointWorld passes degraded actions (wrong conditioning). The expected ordering is V-JEPA2-AC (correct actions) $\geq$ PointWorld (degraded actions) $\geq$ V-JEPA2-No-AC (zero actions), with V-JEPA2-AC providing the strongest discrimination.

Table 3.2 World model families and their methodological roles in the benchmark.

Family	Pretraining	Modality	Role
V-JEPA2-AC	Large-scale video	RGB video	Primary pretrained baseline
LeWM	None (scratch)	RGB video	Provenance-clean from-scratch
DexWM	None (scratch)*	RGB video	Domain-specific (hand-geometry loss)
V-JEPA2 (no AC)	Large-scale video	RGB video	Action-conditioning ablation
PointWorld	Large-scale video	RGB video†	Action-specificity negative control

* The published DexWM architecture targets pretraining on EgoDex and DROID; publicly available weights could not be sourced and the model used here was trained from scratch (Section 3.7).
† The original PointWorld architecture ingests RGB-D input as 3D point clouds. In this study, PointWorld is instantiated as a proxy using the V-JEPA2-AC visual backbone with degraded 7-dimensional robot-arm-style action conditioning; the observation modality is therefore RGB video, not point cloud.

The modality column reports each family's primary input modality. The four RGB video families do not predict or reconstruct pixels: as described in Section 2.3.2 of the literature review, all JEPA-family and encoder-based architectures translate the RGB input into a learned latent representation and perform all prediction and scoring operations in that latent space.

Together, the five families allow comparisons along three independent axes:

pretraining scale (V-JEPA2-AC vs LeWM), action conditioning quality (V-JEPA2-AC vs PointWorld vs V-JEPA2-No-AC, spanning correct to degraded to absent), and domain architecture (V-JEPA2-AC vs DexWM). No single family can be claimed responsible for a result that generalises across all three axes.

3.5 Fair Experimental Protocol

Comparing the five world model families and two policy architectures against each other requires holding every non-design variable constant. Two aspects of fairness are addressed: training exposure and evaluation contract.

3.5.1 Training exposure budget

Different model families have different computational costs per training step and different sensitivities to data volume. Without a principled exposure budget, a comparison could conflate the effect of architecture with the effect of receiving more or fewer training samples. To determine an appropriate shared budget, a convergence study was conducted prior to full training: each model family was trained independently at six data scales (1k, 5k, 10k, 25k, 50k, and 100k effective items) on a fixed subset of the canonical training data, and the resulting loss curves were examined for plateau behaviour. The study and its results are described in detail in Chapter 4; the finding relevant to the experimental design is summarised here.

The effective training exposure E is defined as:

$$E = S \times B \times G \times A \tag{3.1}$$

where S is the number of optimisation steps, B is the per-device batch size, G is the number of gradient-accumulation steps, and A is the number of accelerator devices. This formulation counts the number of training items consumed regardless of how steps are distributed across hardware configurations, allowing models with different batch sizes and GPU counts to be normalised to a common scale.

The convergence study, described in Chapter 4, established a universal budget of $E = 100{,}000$ effective items. This figure is applied uniformly across all model families and all three training splits, ensuring that any measured performance difference reflects architecture and data rather than exposure volume.

3.5.2 Evaluation contract

All models receive the same RGB input at evaluation time: the observation frames from the canonical test window, without additional depth, point cloud, or semantic metadata. This ensures that any performance difference between world model families reflects their internal representation and prediction quality rather than access to a richer input signal. All five families, including PointWorld, operate on the same V-JEPA2 visual latent representations; PointWorld's distinctive characteristic is its degraded action conditioning rather than a different observation modality.

The evaluation windows are fixed at indexing time and shared across all branches and all world model families. No window is selected, filtered, or re-sampled after training. Policy weights and world model weights are frozen before evaluation begins and are not updated. The only variable at evaluation time is the selection rule applied to the candidate pool, isolating the contribution of each branch design and each world model family from all other sources of variance.

3.6 Evaluation Metrics

3.6.1 Offline trajectory prediction setting

EgoDex does not define an official offline evaluation metric. The original benchmark measures policy effectiveness via online task success rate on a physical robot deployment [44], which is outside the scope of this study. The evaluation setting here differs fundamentally: rather than measuring task execution, the study evaluates offline visuomotor trajectory prediction, measuring how closely each configuration's predicted wrist pose sequences align with ground-truth demonstrated trajectories in the held-out test set. Results should therefore be interpreted as trajectory prediction quality under each branch and world model configuration, not as claims about downstream task success. The relationship between offline trajectory prediction quality and online task performance is addressed in Section 2.4 of the literature review and in Section 3.7.

3.6.2 Primary metrics

The action representation throughout this study is a sequence of $\mathrm{SE}(3)$ wrist transforms for both hands (see Section 3.1). Evaluation metrics are defined over this representation directly, decomposed into translation and rotation components.

Translation L2 error. For each predicted action chunk, the Euclidean distance between the predicted and ground-truth wrist positions is computed at each of the 16 prediction steps and averaged across the chunk and across both hands. This yields a

mean per-step translation error in metres, providing an interpretable measure of spatial accuracy over the prediction horizon. Translation L2 is used as the primary evaluation metric in related offline benchmarks [7, 8].

Rotation error. The geodesic distance between the predicted and ground-truth wrist orientations is computed at each step, expressed in degrees. As with translation, values are averaged across the chunk and across both hands. Geodesic distance on $\mathrm{SO}(3)$ is used in preference to Euler-angle difference, which is sensitive to gimbal singularities and axis ordering [44].

The following supplementary metrics are computed alongside the two primaries and reported for completeness. All are expressed as pairwise win rates against the B1 baseline unless stated otherwise.

- **Rotation win rate.** Fraction of windows where the selected candidate achieves lower rotation error than B1; the rotational counterpart to the translation win rate.
- **Endpoint accuracy win rate.** Win rate computed over the mean translation error of the final four prediction steps only, capturing whether selection quality holds through to the end of the prediction horizon.
- **Velocity win rate.** Fraction of windows where the selected candidate's step-to-step velocity profile (first-difference of wrist positions) is closer to the ground-truth velocity profile than B1's selection.
- **Acceleration win rate.** Same as velocity win rate, computed on the second-difference (acceleration) of the wrist position sequence.
- **Straightness win rate.** Fraction of windows where the selected candidate's trajectory is geometrically closer to the ground-truth path shape, penalising unnecessary curvature.
- **Oversmoothing win rate.** Fraction of windows where the selected candidate avoids excessive temporal smoothing relative to the demonstrated trajectory; complements straightness by penalising insufficient motion variation.
- **Nonlinear acceptance rate (B2 only).** Fraction of test windows in which the B2 veto gate fires, i.e. the world model's preferred candidate differs from the policy's and its margin exceeds the threshold. This is a diagnostic for B2 rather than a quality metric; it characterises how often the world model actively intervenes.

3.6.3 Candidate selection comparison

Comparing headline metric values directly across world model families is complicated by the fact that the five families produce scores on different numerical scales, and that absolute metric differences may reflect policy rather than world model quality. To support clean cross-family comparison, candidate selection is also evaluated via a pairwise win rate: for each test window, the candidate selected by a given branch or world model family is compared against the candidate selected by B1 (policy-only), and a win is recorded if the selected candidate achieves lower translation L2 on the ground-truth trajectory; translation L2 is used as the win criterion because it provides the most spatially direct and interpretable measure of selection quality across the 16-step prediction horizon. Win rates are averaged across all test windows (Table 3.1) and reported as percentages. This measure is scale-free and directly answers the question of whether world-model involvement produces a better selection decision than the policy alone, independent of the absolute metric level. The offline evaluation literature supports pairwise comparison as a robust ranking method [43].

3.7 Evaluation Scope and Limitations

This section collects the scope constraints and known limitations of the experimental design. Several have been introduced at relevant points in earlier sections; they are gathered here to give a complete account.

Offline evaluation only. The benchmark is entirely offline: no physical robot is used, and no task completion signal is available. All metrics are computed against ground-truth demonstrated trajectories in the held-out test set. The consequence is that a configuration may achieve strong trajectory prediction scores while still failing at the physical task, and vice versa. The literature reviewed in Section 2.4 establishes that offline rankings can carry predictive validity for physical outcomes when the evaluation domain is well-calibrated to the deployment setting, but this transfer is not guaranteed. The results of this study should be read as evidence about candidate selection quality in the offline trajectory prediction setting, not as claims about downstream robot task success.

Kinematic action space. The action representation is limited to SE(3) wrist pose transforms for both hands: nine dimensions per wrist (a six-dimensional rotation encoding and three translational components), eighteen dimensions in total across the bimanual system. Finger-level articulation is not represented. The evaluation therefore measures bimanual wrist trajectory quality; conclusions do not extend to finger-level dexterous control or to tasks where outcome depends on fine-grained in-hand

manipulation.

No contact or force signals. EgoDex provides RGB video and wrist pose annotations; no contact, force, or haptic signals are present. The action representation is purely kinematic. Tasks for which contact dynamics are the determining factor cannot be evaluated in this setting.

Kinematic domain gap. Demonstrations are produced by human subjects, not a robotic system. The wrist kinematics of a human arm differ from those of a robot manipulator in range of motion, joint compliance, and embodiment. Any downstream robotic deployment would require bridging this domain gap, which is not addressed by the offline benchmark.

Training budget as a compute-constrained lower bound. As noted in Section 3.5.1, the universal training budget of 100k effective items was determined empirically but is not a saturation point for all model families. Some families were still improving at this scale. Performance differences between families may partly reflect differing distances from their respective saturation budgets rather than intrinsic architectural differences.

DexWM trained from scratch. The published DexWM architecture [33] does not provide publicly available model weights: the associated repository contains only trajectory data. As documented in Section 4.5.3, the pretrained checkpoint could not be sourced and DexWM was therefore trained from scratch on the benchmark's declared training splits, following the same procedure as LeWM. This means DexWM results reported here are provenance-clean with respect to the test split, but the trained model does not carry the domain-specific pretraining on EgoDex and DROID that the published architecture was designed to exploit. Results for this family should be interpreted accordingly.

4 Implementation

This chapter describes the concrete implementation decisions made to realise the experimental design specified in Chapter 3. Nine concerns structure the account:

- repository layout and execution model (Section 4.1);
- data pipeline and windowing infrastructure (Section 4.2);
- consolidated training configuration for all model families (Section 4.3);
- policy training, ACT and Diffusion Policy (Section 4.4);
- world model training, all five families (Section 4.5);
- convergence study and universal training budget (Section 4.6);
- B2 margin threshold calibration and guard evaluation infrastructure, together with the fairness safeguards built into it (Sections 4.7–4.9);
- implementation challenges that required significant diagnosis and correction (Section 4.10);
- auxiliary modules excluded from the primary benchmark (Section 4.11).

4.1 Repository and execution model

The research stack is organised as a single Python package under `src/`, with subsystem entry points exposed under `scripts/`. The canonical shell surface follows a per-model naming convention: `policy.sh`, `jepa_wm.sh`, `lewm.sh`, `dexwm.sh`, and one `<model>.sh` per additional model family, complemented by the higher-level orchestration wrapper `run_stage.sh`. Docker is the canonical runtime surface for all stages: datasets, caches, and run outputs are kept outside the repository checkout so that code, data, and generated artefacts remain separated. All reported results are outputs of this staged execution model, tied to frozen run artefacts rather than to manually reconstructed notebook states.

The run output directory follows a split-first, then model layout. Each training job writes its artefacts under `fair_training/<split>/<model>/`, which includes a `result.json` recording the completion status, wall time, and final metric. This layout makes it straightforward to audit which combinations have completed and to compare results across splits for the data-scaling analysis.

4.2 Data pipeline and windowing

4.2.1 Index construction

The index stage is the foundation of the entire stack. It discovers valid paired EgoDex episodes, requiring both a `.mp4` video file and a `.hdf5` pose annotation file, infers episode lengths from the Hierarchical Data Format 5 (HDF5) metadata, and writes canonical observation and prediction windows. The fixed-stride index is configured as `obs16_pred16_s128` following the window protocol described in Section 3.1.2: 16-frame observation windows followed by 16-frame prediction windows, sampled every 128 frames. All training, world model extraction, and guard evaluation in this study consume this windowing contract. The index stage writes resumable checkpoints to allow interrupted builds to continue from the last completed episode.

[Writing note: If Strategy 2 results are incorporated, restructure this section to restore the Strategy 1 / Strategy 2 bold-paragraph pattern. Add a second paragraph describing the state-change-targeted index (peak-detection, task-stratified sampling, one window per episode with a fixed seed, no temporal overlap). Update the opening sentence to note that the pipeline was run separately for each strategy. Update the isolation-property paragraph below to clarify that both strategies enforce the shared-window invariant independently.]

As established in Section 3.2, the shared window definition ensures any measured performance difference is attributable to branch design or architecture rather than to differences in the data each component received.

4.2.2 Incremental training curriculum

Three cumulative training splits were assembled from the first three parts of the EgoDex release, as detailed in Table 3.1 of the methodology. Each split is a strict superset of the previous: the larger splits were assembled using relative symbolic links rather than copying data, allowing incremental construction without duplicating approximately 1 TB of earlier data on disk.

This curriculum allows the data-scaling axis of the benchmark to be evaluated: all five world model families and both policy architectures are trained independently on each of the three splits at the universal budget of 100,000 effective items (Section 4.6), enabling a direct measure of how additional training data diversity affects prediction quality independent of training exposure volume.

4.3 Training configuration

Table 4.1 consolidates the hardware and batch configuration for all model families at the universal budget $E = 100{,}000$. Settings marked *(shared)* are kept identical across all

trainable models. The effective-item budget is verified for each trainable family via $E = S \times B_{\text{global}}$, where S is the number of optimisation steps and B_{global} is the effective batch per step. V-JEPA2 (no AC) and PointWorld require no model training and reuse the Phase 1 extraction shared with V-JEPA2-AC.

Table 4.1 Training configuration for all model families at $E = 100{,}000$. Approximate step time is shown for trainable phases only; Phase 1 of V-JEPA2-AC is a one-time offline extraction with no optimiser step. V-JEPA2 (no AC) and PointWorld reuse the V-JEPA2-AC Phase 1 embeddings and require no additional training (––– denotes not applicable).

Family	**GPUs**	**Batch/device**	**Steps** S	B_{global}	**s/step**
ACT	2×A100	32	1,562	64	≈0.5
Diffusion Policy	1×A100	32	3,125	32	≈2.5
V-JEPA2-AC Ph. 1 (extraction)†	4×A100	—	—	—	—
V-JEPA2-AC Ph. 2 (AC head)	4×A100	4	6,250	16	<0.01
LeWM	4×A100	4 (grad×4)	6,250	16	≈2.5
DexWM	4×A100	4	6,250	16	≈25
V-JEPA2 (no AC)	—	—	—	—	—
PointWorld	—	—	—	—	—

† Phase 1 is a one-time offline extraction (4×A100, sharded, 27.4 windows/s, no backward pass). Measured wall times: `train_part1` ≈0.75 h, `train_part1_2` ≈1.4 h, `train_part1_2_3` ≈2.2 h. Phase 2 head training completes in under ten seconds per split (≈0.01 s/step on cached embeddings). DexWM total wall time across all three training splits was approximately 140 h (≈25 s/step × 6,250 steps × 3 splits); LeWM total wall time was approximately 13 h (≈2.5 s/step, with substantially lower GPU memory footprint than DexWM despite identical step count).

4.4 Policy training

Both policy families share the same observation encoder: a ResNet-18 visual backbone [47] processes the single observation RGB frame (96×96 pixels) through four residual stages (64, 128, 256, and 512 channels), producing a 512-dimensional feature vector that is concatenated with the bimanual wrist state before being passed to the policy-specific decoder (Figure 4.1). Image resizing to 96×96 is applied at loading time and is consistent across all branches. Wrist pose sequences are normalised to zero mean and unit variance using statistics computed from the training split; these statistics are frozen after training and consumed by the world model scoring stages to ensure consistency.

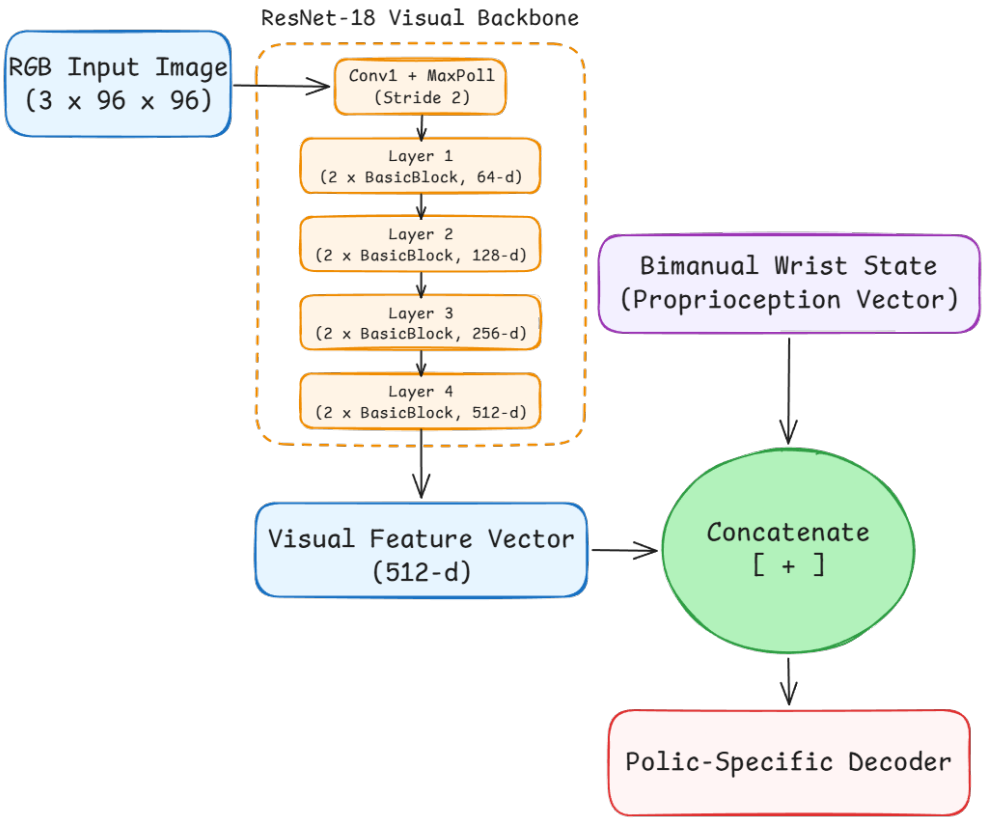


Figure 4.1 Shared ResNet-18 visual backbone preceding both policy decoders. Adapted from [47].

Action Chunking Transformer

The ACT architecture is reviewed in Section 2.1 and illustrated in Figure 2.1. The implementation follows the architecture and training procedure of Zhao et al. [7] with no dependency on an external codebase. Custom components include the EgoDex HDF5 window loader, the 18-dimensional bimanual wrist action space adapter, and a checkpoint serialisation layer that freezes the CVAE configuration and normalisation statistics alongside the model weights. The CVAE encoder-decoder and transformer decoder are realised directly from the paper specification. At training time, the CVAE encoder compresses a ground-truth action chunk together with the current observation into a style variable $z \sim \mathcal{N}(\mu, \sigma^2)$; the transformer decoder reconstructs the chunk from z and the observation. The encoder is discarded at evaluation time, and z is drawn from the standard Gaussian prior $\mathcal{N}(0, I)$. The design intent is to generate up to N diverse candidates by sampling independent z vectors and decoding each under the same observation. Training uses two A100 GPUs with a per-device batch of 32 (global batch 64): $E = 1{,}562 \times 64 \approx 100{,}000$ (Table 4.1). Each run writes a `best.pt` checkpoint selected by lowest validation loss, a `last.pt` checkpoint, per-epoch metrics in `metrics.jsonl`, and a frozen `manifest.json` recording the configuration and window file consumed.

CVAE posterior collapse: investigation and retraining. The trained ACT checkpoint exhibits CVAE posterior collapse: at inference time the decoder is latent-insensitive, and sampling five independent z vectors from $\mathcal{N}(0, I)$ produces five identical action chunks. A diagnostic confirms the extent of the collapse: injecting $z = 10\sigma$ (ten standard deviations from the prior) via a forward pre-hook on the latent projection layer shifts the output by approximately 3×10^{-7} in absolute value, which is effectively zero. The mean pairwise difference across five independently drawn candidates is 0.000000 on all three training splits.

This is a well-documented failure mode of conditional VAE models rather than a mis-implementation [48]. When the conditioning signal (the visual observation) is sufficiently informative to reconstruct the target without consulting z, the decoder learns to ignore z entirely. The KL divergence term in the training objective regularises the *encoder* distribution toward the prior but cannot compel the *decoder* to use the latent; the collapse is therefore not resolved by increasing the KL weight. Three retraining runs were conducted to test this: *v1* used the default `kl_weight=10`; *v2* raised it to `kl_weight=100`, at which point the KL term dominated the total loss (KL ≈ 0.25 nats against a reconstruction loss of ~ 15), yet collapse persisted; *v3* added cyclical KL annealing following Fu et al. [49] (`kl_cycle_batches=200`, `kl_cycle_ramp_frac=0.5`) together with 30 % observation-image dropout to force the decoder to rely on z when visual context is withheld. All three runs produced mean

pairwise differences indistinguishable from zero on every split (Table 4.2). The EgoDex visual observations are sufficiently informative that the decoder finds a loss minimum that ignores z regardless of training pressure, and correcting the collapse without an architectural change (discrete latents, vector-quantisation, or an explicit diversity head) is outside the scope of this thesis.

Table 4.2 ACT CVAE posterior collapse: three successive retraining attempts. `kl_weight`, cyclical annealing schedule, and observation dropout were varied in sequence; all runs produced a mean pairwise candidate difference of zero on every training split.

Run	`kl_weight`	Cyclical KL	Obs. dropout	Mean pairwise diff
v1	10 (default)	no	0 %	0.000000
v2	100	no	0 %	0.000000
v3	100	yes (200-batch cycle, 50 % ramp)	30 %	0.000000

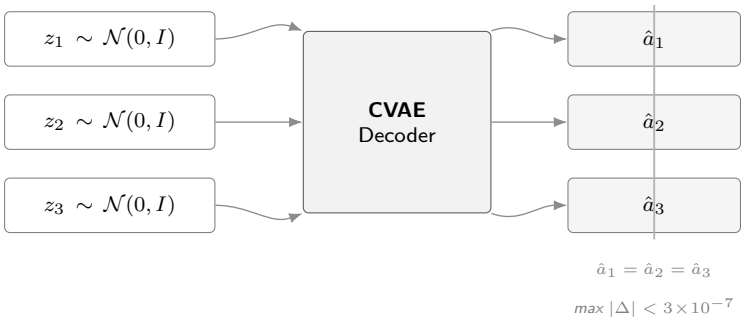


Figure 4.2 CVAE posterior collapse in the trained ACT checkpoint. Three independently sampled latent vectors produce identical outputs; the decoder is latent-insensitive at inference time.

The collapse is treated as an empirical finding. It motivates the use of an external candidate generator, described below, and is reported in Section **??** as a characteristic of the ACT policy on this dataset.

CEMP candidate augmentation. To restore the five-candidate evaluation pool required by all three branches, ACT-based guards use the CEMP planner to generate four alternative action chunks per evaluation window. CEMP is an adaptation of the Cross-Entropy Method [50] initialised from the policy's own output rather than a uniform prior. It operates entirely in action space and is independent of the ACT model weights; the collapsed latent is never consulted. Concretely, CEMP takes the ACT Branch 1 output as a base chunk and runs three optimisation iterations. In each iteration, 64 candidate chunks are drawn by perturbing the current mean in a low-dimensional temporal basis space: the 18-dimensional action is decomposed into semantic groups (three translation and six rotation dimensions per wrist), and

perturbations for each group are parameterised as a linear combination of four temporal basis functions (constant, linear, quadratic, and sinusoidal). This `basis_groups` parameterisation constrains search to smooth, temporally coherent trajectories rather than independent per-timestep noise, which is appropriate for bimanual wrist motion. The 64 candidates are scored by the world model's latent prediction function; the top 25 % (16 elites) define the next iteration's mean and variance, with the standard deviation decaying by a factor of 0.7 per iteration. After three iterations the four highest-scoring candidates across all iterations are returned. The scoring function used inside CEMP is the same world model that performs the final selection; this is standard practice in model-based planning (following TD-MPC2 [37] and similar) and means the CEMP candidates are already pre-filtered toward actions the world model prefers. The final selection step is consequently a form of argmax confirmation rather than an independent evaluation; the B2 and B3 results should be interpreted as measuring the quality of the combined world-model-plus-planner system, not the world model's scoring ability in isolation.

For Branch 2 guards the pool contains these four CEMP alternatives plus the ACT Branch 1 output, forming a five-candidate pool with the policy's own proposal included. For Branch 3 guards the policy output is excluded, and the pool contains only the four CEMP alternatives. This design preserves the methodological intent of the three-branch framework: B1 represents the unaided policy, B2 tests whether a world model can identify a superior action from a pool that includes the policy's own proposal, and B3 tests whether a world model can generate a superior action independently of the policy.

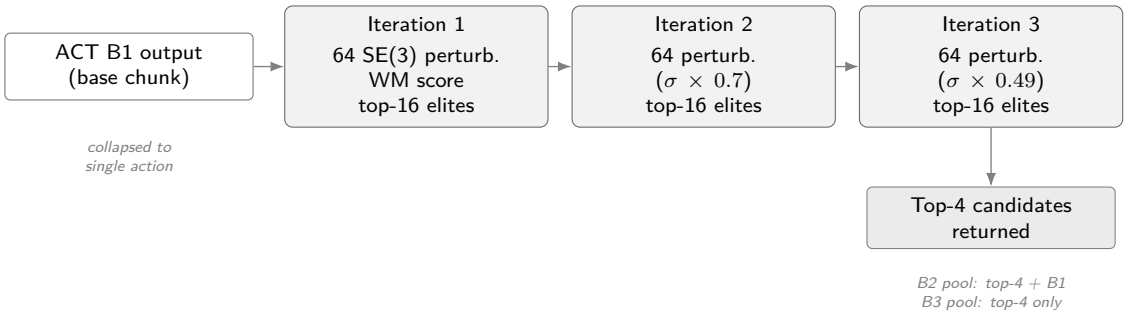


Figure 4.3 CEMP candidate augmentation for ACT-based guards. Three optimisation iterations refine perturbations around the policy's collapsed output; the standard deviation decays by $0.7\times$ per iteration. The four highest-scoring candidates across iterations form the B2/B3 evaluation pool.

Diffusion Policy

The Diffusion Policy implementation follows the transformer-backbone variant (DP-T) described in Section 2.2 and illustrated in Figure 2.2. The implementation follows Chi

et al. [8] with no external codebase dependency. Custom components include the 18-dimensional bimanual wrist action space adapter, the EgoDex HDF5 window loader, and the five-candidate DDIM evaluation protocol (five independent noise seeds decoded under the shared conditioning signal). The same ResNet-18 observation encoder produces a conditioning vector `obs_cond` passed to a DiT denoising backbone via cross-attention. At evaluation time, five independent DDIM denoising chains are run on the same observation to produce five candidate action chunks, each starting from independently sampled Gaussian noise and converging to a distinct trajectory under the shared conditioning signal. Training uses one A100 GPU with a per-device batch of 32: $E = 3{,}125 \times 32 = 100{,}000$ (Table 4.1). The same checkpoint, metrics, and manifest artefacts as ACT are written at the end of each run.

DDIM subsampling correction. An implementation audit conducted after the initial evaluation runs identified an error in the reverse-diffusion sampler. The DDIM step (Song et al. [46]) requires the cumulative product $\bar{\alpha}_{t_{\text{prev}}}$ where t_{prev} is the *previous timestep in the subsampled inference schedule*. The original implementation passed $\bar{\alpha}_{t-1}$ (the consecutive timestep in the training schedule) at every step, so a ten-step schedule spanning $t \in \{90, 80, \ldots, 0\}$ applied $\bar{\alpha}_{89}$ at the first step rather than the correct $\bar{\alpha}_{80}$. This miscalibrates the predicted $\hat{x}_0$ weighting at each reverse step and marginally degrades sample quality, but does not affect the diversity guarantee: each of the five candidates still starts from an independently sampled $x_T \sim \mathcal{N}(0, I)$ and the collapsed-latent failure mode present in ACT cannot arise here. The fix (commit `92e8116`) passes the explicit next-in-schedule index to `ddim_step` and re-runs all three evaluation splits under the corrected sampler; the corrected results supersede the initial runs.

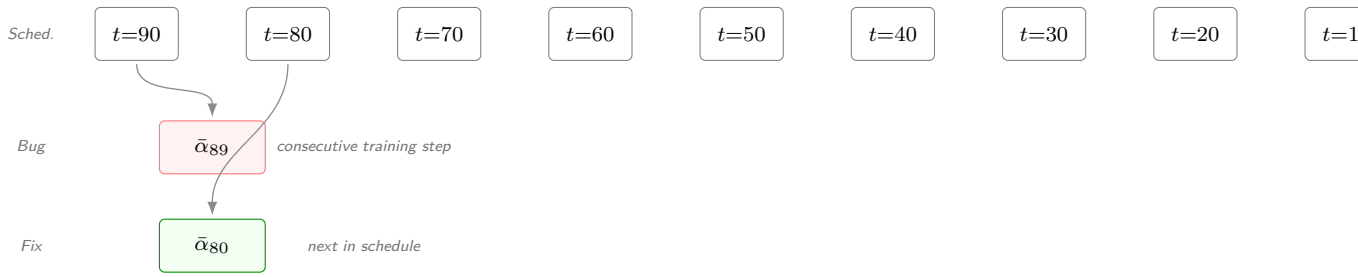


Figure 4.4 DDIM subsampling correction. The ten-step inference schedule spans $t \in \{90, 80, \ldots, 0\}$. At each reverse step the correct $\bar{\alpha}_{t_{\text{prev}}}$ is the *next timestep in the subsampled schedule*; the bug substituted the consecutive training-schedule timestep $\bar{\alpha}_{t-1}$, miscalibrating the $\hat{x}_0$ weighting at every step.

4.5 World model training

4.5.1 V-JEPA2-AC: extraction and transition head

The V-JEPA2-AC architecture is reviewed in Section 2.3.2. Figure 2.6 illustrates the standard V-JEPA2 backbone alongside the action-conditioned variant that forms Phase 2. The frozen ViT-Large backbone is loaded from the `facebook/vjepa2` HuggingFace repository and cached locally; the action-conditioned transition head is implemented and trained from scratch. Training separates into two phases.

In Phase 1, the frozen ViT-Large encoder processes all indexed windows in a single batched inference pass across four A100 GPUs. The encoder takes 16-frame observation sequences and produces a latent embedding vector per window; these are written to `embeddings.npy` together with a companion `window_ids.txt` recording the exact identifier (split, task, episode, t_0) of each row. Phase 1 is a one-time offline extraction: there is no backward pass, no optimiser, and no epoch loop. A critical bug in an early run applied a 2,000-window default cap producing only approximately 1,000 training pairs from a split of 140,000 windows; this is documented in Section 4.10.

In Phase 2, a small MLP transition head is trained on the stored embeddings to predict the next embedding z_{t+1} from the current embedding z_t and an action conditioning vector derived from the bimanual wrist state delta (`action_repr=policy_state_delta`). Training runs on four A100 GPUs: $E = 6{,}250 \times 16 = 100{,}000$ (Table 4.1). Wall-clock time is under ten seconds because each step is a small MLP forward–backward pass with no backbone inference. The trained head is exported via an `act_adapter.json` contract loaded by the guard stage at runtime.

4.5.2 LeWM: end-to-end training

LeWM is reviewed in Section 2.3.2 and illustrated in Figure 2.8. The architecture is implemented following the published specification [34] and trained entirely from scratch with no external weights. Training jointly optimises the ViT-Tiny encoder, the autoregressive predictor, the action embedder, and the projector from random initialisation using the next-embedding prediction loss and the SIGReg regulariser (Equation 2.1) that enforces a Gaussian prior on the latent distribution [34].

EgoDex episodes are exported into contiguous HDF5 sequences containing pixel frames, per-step bimanual wrist state, per-step action (state delta), and per-episode length metadata. The frame cap is derived from the maximum window start position across all benchmark splits: the test split requires coverage up to frame 5,152 and `train_part1_2` up to frame 5,280, so the export contract adopts

`max_frames_per_episode=5280` throughout. An earlier cap of 256 frames caused 97.9 per cent of test windows to be silently excluded; the full resolution of this defect is documented in Section 4.10.

LeWM training runs on four A100 GPUs with a per-device batch of 4 and four gradient-accumulation steps (global effective batch 16): $E = 6{,}250 \times 16 = 100{,}000$ (Table 4.1). Wall-clock training time on `train_part1_2_3` was approximately 4.3 hours per run (≈2.5 s/step $\times$ 6,250 steps), for a total of approximately 13 hours across all three training splits.

A normalisation statistics defect discovered during the evaluation audit caused the guard stage to load `action_mean` and `action_std` from the test-set export rather than the training-set export, introducing a 1.74$\times$ scale mismatch in the LeWM cost function across all initial guard runs. The full description and resolution are documented in Section 4.10.

4.5.3 DexWM: from-scratch training

DexWM is reviewed in Section 2.3.2. Its pipeline pairs a frozen DINOv2 image encoder with a CDiT predictor and incorporates a hand-consistency loss that supervises fingertip and wrist heatmap predictions (Figure 4.5). The DINOv2 ViT-Large encoder is loaded from the `facebookresearch/dinov2` HuggingFace repository; the CDiT predictor and HC head are implemented from the published architecture [33] and trained from scratch.

Checkpoint sourcing investigation

The published DexWM architecture [33] does not provide publicly available model weights. The associated HuggingFace repository (`facebook/dexwm`) contains only trajectory data in `.hdf5` format; no weight file is present. The GitHub repository README states that pretraining requires 32 nodes of 8 GPUs each for several days on the EgoDex and DROID datasets, and does not link to a downloadable checkpoint. Automated download attempts confirmed a 404 response for all candidate weight paths. The codebase initially contained a silent fallback that substituted an identity stub (`_CDiTStub`) returning zero scores for all candidates without any user-visible error. This stub was replaced with an explicit hard-failure preventing any experiment from silently using zero-score predictions.

Because no external checkpoint could be sourced, DexWM was trained from scratch on the benchmark's declared training splits, adopting the same provenance stance as LeWM: no external pretraining and no data outside the declared splits. The DINOv2 ViT-Large encoder (1,024-dimensional patch tokens) remains frozen; the CDiT

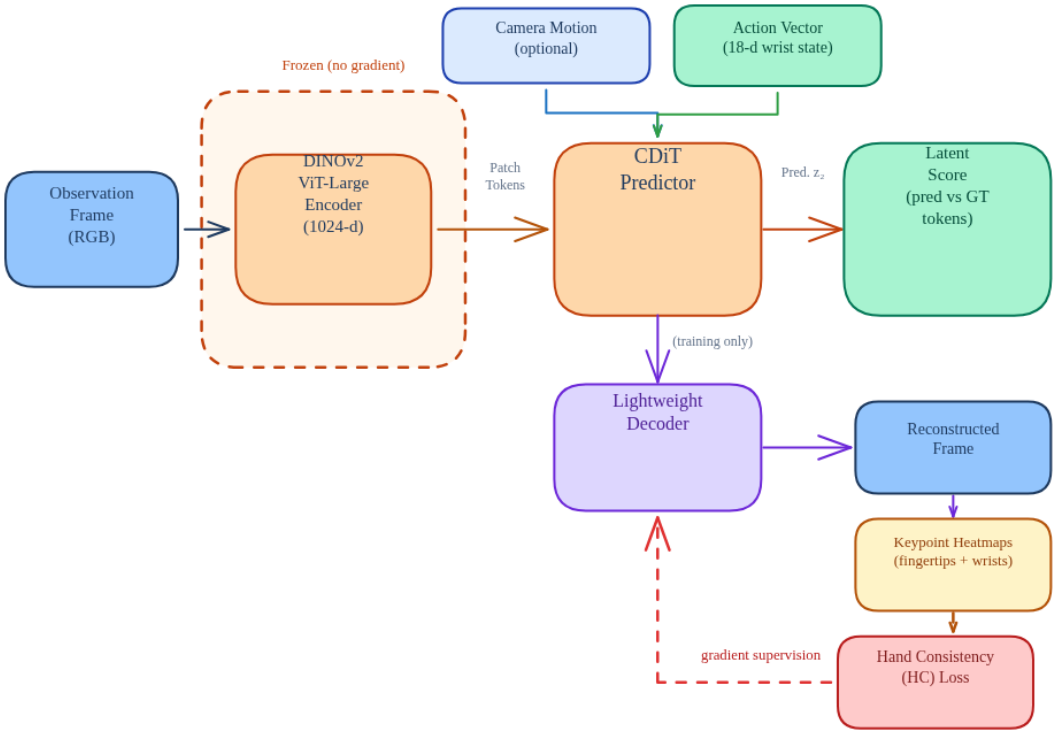


Figure 4.5 DexWM pipeline as implemented, comprising a frozen DINOv2 encoder, CDiT predictor, lightweight decoder, and HC loss supervision path. Adapted from [33].

predictor and HC head are trained from random initialisation. An early run misconfigured the ViT-Base variant (768-dimensional), causing all guard-stage inference to fail silently; see Section 4.10.

DexWM training runs on four A100 GPUs with a per-device batch of 4 (global batch 16): $E = 6{,}250 \times 16 = 100{,}000$ (Table 4.1). Wall-clock training time was approximately 43 hours per split (≈25 s/step), for a total of approximately 140 hours across all three training splits, reflecting the substantially higher GPU memory footprint of the patch-level spatial representations relative to LeWM's global embedding approach.

An evaluation audit subsequently found that the initial generic guard runs for DexWM loaded the ACT Branch 1 output as the sole candidate ($N = 1$), making the world model's scoring a vacuous no-op. The canonical DexWM evaluation uses a dedicated pipeline that draws eight retrieved training demonstrations per window; the full description is in Section 4.10.

4.5.4 V-JEPA2 (no action conditioning): ablation control

The methodological role of this ablation is described in Section 3.4. In implementation, the variant is realised by reusing the Phase 1 embeddings extracted for V-JEPA2-AC; no

additional training is required (Table 4.1). At guard time, the scorer passes a zero action vector to the transition model for all candidates, producing action-blind scores that reflect visual plausibility alone. A dedicated result artefact records the persistence loss (mean MSE plus cosine distance between consecutive frame embeddings, evaluated without action input) as a diagnostic metric confirming that the ablation ran correctly.

4.5.5 PointWorld: action-degradation control

The methodological role of the PointWorld control is described in Section 3.4. The implementation uses the same frozen V-JEPA2-AC backbone and transition head as the V-JEPA2-AC family, requiring no additional model training (Table 4.1). The sole difference is that candidate action chunks are degraded before being passed to the transition head.

A `HandToArmActionAdapter` projects the 18-dimensional bimanual wrist representation (rot6d 6D + xyz 3D per wrist) to a seven-dimensional robot arm representation (six wrist dimensions plus a scalar gripper aperture derived from the mean of remaining dimensions) via the `wrist_plus_gripper` strategy, then zero-pads the result back to 18 dimensions. The right-hand contribution is discarded entirely; the 11 recovered dimensions carry no meaningful hand-motion signal. The transition head therefore receives the correct visual features but systematically corrupted action conditioning.

The convergence study diagnostic additionally computes a proxy loss:

$$\mathcal{L}_{\text{PointWorld-proxy}} = \mathcal{L}_{\text{persist}} + 0.5 \cdot \text{mismatch} \tag{4.1}$$

where $\mathcal{L}_{\text{persist}}$ is the frame-to-frame embedding distance and mismatch is the fractional norm loss of the projection residual. This diagnostic confirms that the adapter introduces genuine information loss: the bimanual hand action space does not map faithfully to the 7-DOF arm representation, as expected for a degradation control.

4.6 Convergence study and universal training budget

As described in Section 3.5.1 of the methodology, a convergence study was conducted prior to full fair training to determine an appropriate shared exposure budget. Each of the five primary model families was trained independently at six data scales (1k, 5k, 10k, 25k, 50k, and 100k effective items) on a fixed stratified subset of `train_part1_2_3`. The two control families (V-JEPA2 no-AC and PointWorld) were included in the sweep but excluded from the budget selection decision.

4.6.1 Loss curves and plateau detection

Table 4.3 reports the validation loss at each scale for all seven families. Each row uses a different training objective so raw values are not comparable across families; the relevant signal is each family's own trend across scales. Bold entries mark the earliest scale at which a family's loss saturates (improvement below 0.5 per cent relative to the next scale).

Table 4.3 Validation loss across six training scales for all seven model families. Bold marks each family's first scale of saturation; losses are objective-specific and not cross-family comparable.

Family	1k	5k	10k	25k	50k	100k
ACT	**0.134**	0.134	0.129	0.133	0.130	0.129
Diffusion	10.86	5.22	4.25	3.11	2.57	2.14
DexWM	4.31	2.41	2.00	1.90	1.69	1.36
LeWM	0.335	0.209	0.165	0.145	**0.130**	0.130
V-JEPA2-AC	1.796	0.375	0.308	0.286	0.257	0.227
V-JEPA2 (no AC)	0.460	0.477	0.373	0.352	0.353	0.377
PointWorld	0.655	0.679	0.628	0.629	0.632	0.592

The bold entries identify the two families that saturate within the sweep: ACT is essentially flat from 1k onwards (a 4 per cent improvement over a 100-fold data increase), and LeWM plateaus between 50k and 100k (0.5 per cent improvement). V-JEPA2-AC, Diffusion Policy, and DexWM carry no bold entry because all three are still declining at 100k, as Figure 4.6 shows. The two controls show no consistent downward trend, consistent with their respective proxy designs having no learnable relationship with additional data.

4.6.2 Universal budget decision

The plateau budget per primary family is summarised in Table 4.4. Given that Diffusion Policy, DexWM, and V-JEPA2-AC all continue to improve at 100k, the universal budget is set to $E = 100{,}000$ as a compute-constrained lower bound. ACT and LeWM, which plateau earlier, receive the same 100k budget by design: the extra exposure cannot hurt them and ensures all families are compared under the most favourable common condition reachable within the compute envelope.

This budget is a compute-constrained lower bound rather than a true saturation point for all families. The performance gaps between families reported in the results chapter may partly reflect differing distances from their respective saturation budgets rather than intrinsic architectural differences. This limitation is acknowledged in Section 3.7 of the methodology.

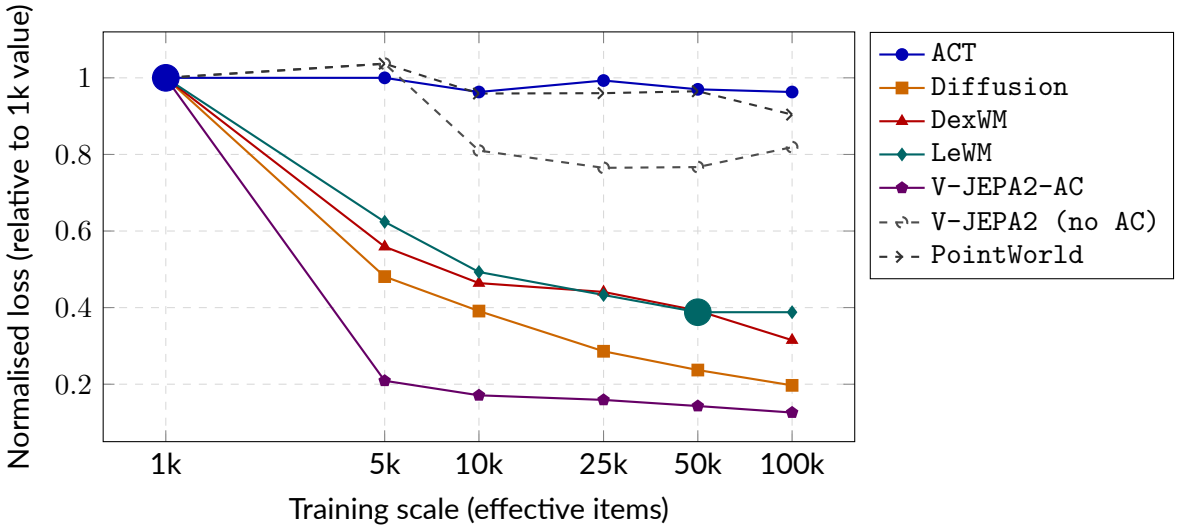


Figure 4.6 Normalised validation loss (each family's value divided by its 1k loss) across six training scales. Solid lines are primary models; dashed lines are controls. Large filled circles mark saturation points matching the bold entries in Table 4.3. Families with no marker are still declining at 100k.

Table 4.4 Plateau budget per primary model family. The universal budget is $E = 100{,}000$ effective items.

Family	**Plateau budget**	**Evidence**
ACT	1k	Flat from 1k; 4% drop over 100× more data
LeWM	50k	50k→100k improvement of 0.5%; effectively converged
V-JEPA2-AC	>100k	11% drop at 50k→100k; still declining
Diffusion	>100k	17% drop at 50k→100k; still declining steeply
DexWM	>100k	20% drop at 50k→100k; still declining
Universal budget		$\boldsymbol{E = 100{,}000}$

4.7 B2 margin threshold calibration

The Branch 2 guard design includes an optional margin threshold: the world model's override is accepted only when its preferred candidate outscores the policy's own output by at least a fixed margin. A threshold of zero causes the world model to override unconditionally on every window where it prefers a different candidate; a high threshold suppresses most overrides and collapses B2 toward B1. To understand this trade-off, a development sweep was run on the validation split of `train_part1_2` using retrieval-based candidate pools (nearest-neighbour training demonstrations as alternatives). The swept margins were $\{0.25, 0.50, 1.00\}$ for the V-JEPA2-AC guard. The results illustrated a clear monotone pattern: lower margins produced higher override rates and lower immediate loss ratios relative to B1, while higher margins suppressed overrides to near-zero with correspondingly small loss improvements. At margin 0.25 approximately 21 per cent of windows were overridden; at 1.0 the override rate fell below 0.2 per cent, recovering only 6 per cent of any potential gain.

The design decision following this sweep was to adopt unconditional argmax (`override_margin=0.0`) for the canonical evaluation. Two considerations drove this choice. First, when the candidate source was migrated from retrieval to CEMP (Section 4.9), the threshold question changed: CEMP already pre-filters candidates toward actions the world model prefers through its internal scoring loop, making a post-hoc margin gate partially redundant. Second, unconditional argmax provides a cleaner experimental signal: the world model's discriminative quality is measured directly, without a threshold hyperparameter that would require separate calibration per world model family and per candidate pool type. All canonical guard runs in this study therefore use `selection_mode: best_score` with `override_margin=0.0`. The development sweep results are retained for transparency but are not used to derive any reported result.

DexWM was excluded from the development sweep. The initial generic DexWM guard runs were subsequently found to be vacuous due to a candidate-loading defect (Section 4.10); the canonical DexWM evaluation uses the dedicated `guards_native_dexwm/` pipeline described in Section 4.8.

4.8 Guard evaluation

4.8.1 Branch 1: policy-only baseline

Branch 1 records the candidate that the policy scores highest by its own internal preference for each test window. For ACT, the CVAE posterior collapse described in

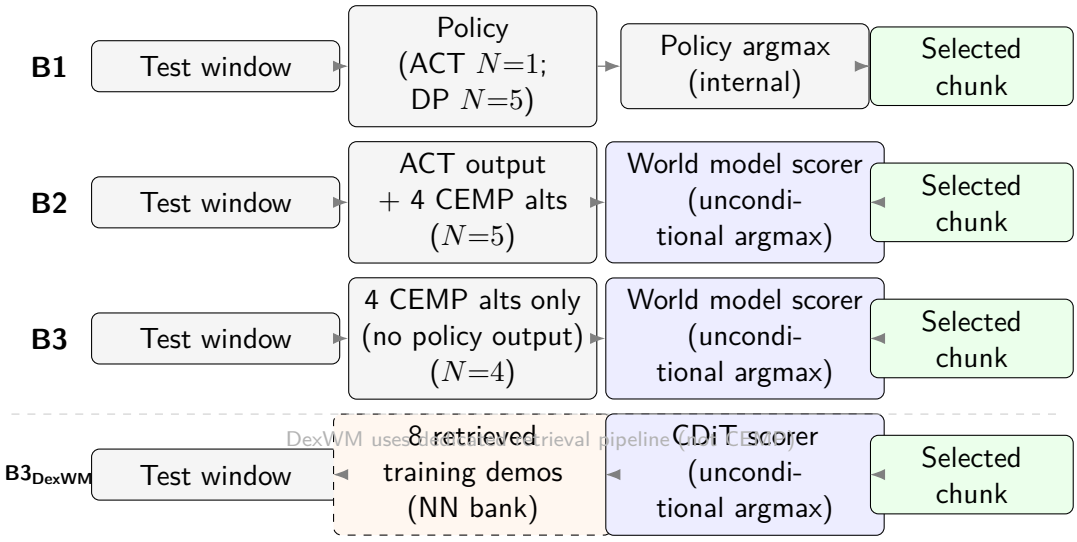


Figure 4.7 Guard evaluation pipeline for all branches. B2 pools include the ACT output alongside four CEMP-planned alternatives; B3 pools contain the four CEMP alternatives only, with the policy output excluded. The DexWM B3 variant (dashed border) uses a distinct pipeline in which eight retrieved training demonstrations replace the CEMP pool; see Section 4.8.

Section 4.4 means a single deterministic output is produced; Branch 1 records that output directly. For Diffusion Policy, Branch 1 selects the candidate with the lowest denoising residual among the five independent chains. No world model is queried.

4.8.2 Branch 2: world-model-inclusive selection

The Branch 2 guard scores a five-candidate evaluation pool per window using the latent prediction pipeline of the relevant world model family. For ACT-based guards the pool contains the ACT output from Branch 1 alongside four alternatives generated by a CEMP planner initialised from the policy's action distribution. The world model scores all five candidates and selects the highest-scoring one unconditionally (`selection_mode: best_score`); no margin threshold is applied. The ACT output is retained only if the world model assigns it the highest score.

For V-JEPA2 without action conditioning the world model cannot produce candidate-specific predictions, so the policy output scores highest by default and Branch 2 reduces to Branch 1 for that model — an expected null result.

The same guard structure is applied independently to each world model family. DexWM uses its CDiT predictor operating on DINOv2 patch tokens rather than a single global embedding. LeWM uses its ViT-Tiny encoder and autoregressive predictor; the guard encodes a rollout prefix to establish latent context before scoring. The PointWorld guard uses its proxy scoring function described in Section 4.5.5.

An evaluation audit (Section 4.10) found that V-JEPA2-AC, V-JEPA2 (no AC),

and PointWorld guards initially used retrieval-based candidate pools rather than CEMP. All three families were re-evaluated with `candidate_source=planner` to ensure a consistent candidate source across all world model families; the retrieval-based runs are archived and not reported.

4.8.3 Branch 3: world-model-exclusive selection

Branch 3 uses a pool of four CEMP-planned alternatives per window with the policy's own output absent (`selection_mode: world_model_only`). The world model applies the same scoring procedure as Branch 2 and selects the highest-scoring candidate unconditionally. The correctness of this design is verified through the guard manifest: every B3 run should record all 7,607 windows as `override_reason_counts: {world_model_only_best_non_act: 7607}`, confirming the policy output is absent from the pool. This manifest invariant was used to detect and correct the Branch 3 selection-mode error documented in Section 4.10.

DexWM is an exception to this frozen-pool design. Its Branch 3 evaluation does not score policy-generated candidates; instead, candidates are drawn from the nearest-neighbour training bank described in Section 4.5.3. The CDiT model scores eight retrieved training demonstrations per window and selects the best without any policy action present in the candidate set. This constitutes a strictly stronger Branch 3 claim: the world model must be sufficient for action selection independently of the policy. DexWM B3 is therefore evaluated separately via a dedicated pipeline (`run_dexwm_native_eval_docker.sh`), and its results are reported alongside but distinguished from the generic B3 results for the other four world model families.

4.8.4 Manifest and checkpoint design

Every guard run emits a frozen manifest at completion recording the selection mode, candidate source, aggregate score statistics, and per-reason override counts. Guard runs also write per-window JSONL checkpoints flushed after each window, allowing interrupted runs to be resumed from the last completed window via a `--resume-dir` flag rather than restarted from scratch.

4.9 Evaluation safeguards

Oracle goal elimination. In early experiments, the latent goal used for scoring was derived from the ground-truth future frame of each episode, giving Branches 2 and 3 access to information unavailable at deployment time. This was corrected by setting

`goal_source=none` in all canonical benchmark runs. All results in the results chapter use this corrected configuration.

Candidate pool invariant. Both Branch 2 and Branch 3 score five-candidate pools generated independently per window by the CEMP planner. Branch 2 pools contain the ACT output plus four planned alternatives; Branch 3 pools contain four planned alternatives only. The key controlled variable between the two branches is pool membership (policy output present vs. absent), not the number of candidates, which is five for both. The pool-size correctness of every B3 run is verified through the guard manifest: all 7,607 override-reason records must read `world_model_only_best_non_act` with zero policy fallbacks. This invariant was used to detect the Branch 3 selection-mode error documented in Section 4.10.

Shared comparison surface. The `policy_chunk_compare` stage computes all cross-branch and cross-family metrics on the same prediction targets rather than relying on controller-native summaries, ensuring that translation L2 and rotation error values are computed identically across all configurations. The evaluation window-count shortfall documented in Section 4.10 was detected by checking `num_windows` in each manifest against the expected 7,607.

Test-split isolation. The test split was held out throughout all stages of the project. All training, hyperparameter selection, threshold calibration, and qualitative development work used training and validation splits only.

Candidate source standardisation. An audit conducted after the initial evaluation runs found that V-JEPA2-AC, V-JEPA2 (no AC), and PointWorld used retrieval-based candidate pools (`candidate_source=retrieval`) while LeWM used CEMP-planned alternatives (`candidate_source=planner`). Because the candidate source is a controlled variable in the B2/B3 comparison, a source asymmetry confounds cross-model comparisons: differences between families could reflect candidate quality rather than world model quality. All five families were therefore re-run with `candidate_source=planner` before any results were reported. The original retrieval-based runs are archived under `*_retrieval_archived/` subdirectories and are not used in this thesis. The full audit is documented in Section 4.10.

4.10 Implementation challenges and resolutions

Thirteen significant implementation challenges arose during the project and were diagnosed and resolved before the final benchmark results were reported. The first seven concern training and infrastructure; the remaining six emerged during the evaluation audit and affected the canonical guard evaluation results.

V-JEPA2 extraction window cap.

- *Symptom:* Default 2,000-window extraction cap produced approximately 1,000 training pairs from a split of 140,000 windows (95-fold shortfall).
- *Impact:* All V-JEPA2-AC transition head runs under the capped extraction were invalid.
- *Fix:* Added `--full` flag to the extraction entry point; re-runs confirmed results consistent with convergence study predictions.

LeWM CPU inference.

- *Symptom:* Hard-coded `.to("cpu")` in the model-loading code placed the guard model on CPU despite available GPU access.
- *Impact:* Estimated guard run time of approximately 54 hours for 7,607 test windows at five candidates each.
- *Fix:* CUDA availability check at load time; model and all tensors moved to GPU. Throughput improved to approximately 19.5 windows per second, reducing guard run time to under seven minutes.

LeWM episode coverage defect.

- *Symptom:* Initial `max_frames_per_episode=256` export caused windows with start positions beyond frame 256 to be silently excluded.
- *Impact:* First LeWM evaluation covered only 160 of 7,607 intended test windows (2.1 per cent coverage).
- *Fix:* Frame cap computed from actual benchmark window lists (5,280 frames); export and training rebuilt; coverage raised to 100 per cent.

LeWM CPU-only training shortfall.

- *Symptom:* Omission of `--train-accelerator gpu --train-devices 4` flags caused training to run on a single CPU.
- *Impact:* Four-fold effective-item shortfall relative to the intended budget.
- *Fix:* Flags added; all affected runs invalidated and re-run with the correct configuration.

DexWM guard dimension mismatch.

- *Symptom:* DINOv2 ViT-Base (768-dimensional) loaded instead of ViT-Large (1,024-dimensional); guard inference failed with a dimension error.
- *Impact:* Silent fallback stub produced zero scores for all candidates, masking the failure in guard outputs.
- *Fix:* Silent fallback replaced with explicit hard-failure; correct ViT-Large configuration confirmed throughout.

Branch 3 selection mode error.

- *Symptom:* Branch 3 launched without `--selection-mode world_model_only`; default `best_score` mode combined policy and world model scoring.

- *Impact:* Branch 3 produced output identical to Branch 2 across all 7,607 windows; detected via a 0.000 win rate in pairwise comparison.
- *Fix:* Re-ran Branch 3 with correct flag; verified via manifest override count (7,607 world-model decisions, 0 policy fallbacks).

HDF5 file handle exhaustion.

- *Symptom:* LeWM guard opened `episodes.h5` inside the per-window scoring function, producing approximately 38,000 open/close operations per test-set guard run.
- *Impact:* Risk of descriptor exhaustion on systems with low file-handle limits.
- *Fix:* File opened once during runtime initialisation; handle held for the duration of the run.

Diffusion Policy DDIM subsampling error.

- *Symptom:* The DDIM reverse step consumed $\bar{\alpha}_{t-1}$ (consecutive training-schedule index) instead of $\bar{\alpha}_{t_{\text{prev}}}$ (previous index in the subsampled inference schedule). A ten-step schedule spanning $t \in \{90, 80, \ldots, 0\}$ therefore applied $\bar{\alpha}_{89}$ at the first step rather than the correct $\bar{\alpha}_{80}$.
- *Impact:* Miscalibrated $\hat{x}_0$ weighting at every reverse step; marginally degraded sample quality. Candidate diversity was unaffected because each of the five chains still starts from an independently sampled x_T.
- *Fix:* Commit `92e8116` passes the explicit next-in-schedule index to `ddim_step`. All three Diffusion Policy evaluation splits were re-run with the corrected sampler; the corrected results supersede the initial runs. Full description in Section 4.4.

PointWorldScorer stale constructor kwargs.

- *Symptom:* `guard.py` instantiated `PointWorldScorer` with keyword arguments `predictor=` and `latent_dim=` that had been removed in an earlier API refactor, raising a `TypeError` at guard startup.
- *Impact:* All PointWorld guard runs failed to launch until the constructor call was corrected.
- *Fix:* Stale kwargs removed; constructor updated to the current API. Fix included in commit `92e8116`.

`_s_sq_cache` memory-pointer collision.

- *Symptom:* The scorer base class cached score scale factors in a dictionary keyed by the Python object's memory address (`id()`). After garbage collection, a newly allocated tensor could reuse the same address as a previously cached object, causing stale scale factors to be broadcast against the new tensor.
- *Impact:* Incorrect guard scores beginning at approximately window 1,700 of a run, producing silent result corruption without any raised exception.

- *Fix:* Cache keyed on an explicit run-scoped string identifier rather than the object address. Fixed in commit `89e2af6`.

Evaluation window count shortfall.

- *Symptom:* Initial guard launch scripts omitted `--eval-ratio 0.0` and `--max-eval-batches 100000`, causing the evaluation loop to exit after the default 787-window limit.
- *Impact:* Only 787 of the 7,607 intended test windows were evaluated (10.3 per cent coverage); results were statistically unreliable and not representative of the full test set.
- *Fix:* Flags added to all guard launch scripts. Coverage confirmed via `num_windows` in the run manifest against the expected 7,607.

DexWM vacuous single-candidate evaluation.

- *Symptom:* The generic `b2_dexwm` and `b3_dexwm` guard runs loaded the ACT Branch 1 output as the sole candidate ($N = 1$). Because the CVAE is fully collapsed, this candidate is deterministic; the DexWM world model therefore scored a pool of one and its selection was always the trivially forced choice.
- *Impact:* The guard results were equivalent to Branch 1 regardless of the DexWM model quality; the world model's discriminative capacity was never exercised. Detected by inspecting `candidate_count_mean: 1.0` in the run manifest.
- *Fix:* Generic DexWM guard runs archived under `b2_dexwm_archived/` and `b3_dexwm_archived/`. Canonical DexWM evaluation uses the dedicated `guards_native_dexwm/` pipeline, in which Branch 3 draws eight retrieved training demonstrations as candidates and scores them with the CDiT predictor (Section 4.8).

LeWM normalisation statistics from wrong export.

- *Symptom:* The `action_mean` and `action_std` tensors consumed by the LeWM guard at runtime were read from the test-set export rather than the training-set export. The test-set action standard deviation (≈ 0.014) differed from the training-set value (≈ 0.024) by a factor of approximately 1.74.
- *Impact:* The LeWM cost function operated on incorrectly scaled action representations throughout every candidate scoring call, distorting the world model's preference ordering in a dataset-split-dependent way.
- *Fix:* Guard configuration updated to read normalisation statistics from the training export. Fixed in commit `d188816`; all three LeWM guard runs re-executed under the corrected normalisation.

Candidate source asymmetry across world model families.

- *Symptom:* An audit of the initial B2/B3 evaluation runs found that V-JEPA2-AC, V-JEPA2 (no AC), and PointWorld used `candidate_source=retrieval` (expert

demonstration neighbours), while LeWM used `candidate_source=planner` (CEMP-generated alternatives). The candidate source is a controlled variable in the B2/B3 design: it determines what alternatives the world model evaluates, not only which model scores them.

- *Impact:* Cross-family comparisons were confounded: differences between, say, V-JEPA2-AC and LeWM could reflect candidate quality rather than world model quality. The three retrieval-based runs are archived under `*_retrieval_archived/` subdirectories.
- *Fix:* All five world model families re-run with `candidate_source=planner` (CEMP). The standardised results are reported in the results chapter; see also Section 4.9.

4.11 Auxiliary modules

Two auxiliary modules are fully implemented in the repository but were excluded from the final benchmark evaluation.

The semantic alignment stage converts indexed windows into captions, structured tags, and text embeddings. The pipeline uses a Vision Language Model (VLM) to generate free-form descriptions and task-relevant categorical tags for each window, followed by a separate embedding model that maps the text to a shared representation space. A critic-and-retry loop rejects low-quality outputs and requests regenerated captions.

The object grounding and mask generation stage consumes semantic object prompts from the alignment stage and produces bounding box detections and segmentation masks for identified objects. The current implementation provides pixel-space localisation of objects but does not estimate $\mathrm{SE}(3)$ object pose.

Both modules were excluded from the primary benchmark because a shared input contract across all five world model families could not be maintained: the semantic and grounding outputs could not be wired into the frozen DINOv2 and ViT-Large encoder paths used by DexWM, V-JEPA2-AC, and their shared embedding reuse families, violating the fairness requirement established in Section 3.5.2. Their implementation is documented in Appendix **??**.

These modules pre-date the current benchmark scope. They were developed when the initial research direction explored semantic feature enrichment as a pre-training pipeline for world models, before the study evolved toward a comparative evaluation of published architectures across a standardised five-family design. The initial intuition that frozen encoder pre-extraction followed by lightweight head training is a productive approach has since been validated by the architectures of DexWM, DINOv2-based world models, and related state-of-the-art systems, each of

which adopts an analogous two-phase structure. This line of work remains a natural direction for continued research beyond the scope of the current results.

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